

GRAVITATIONAL DUALS FROM EQUATIONS OF STATE

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THE COMPLEXITY OF THE COSMOS

GRAN SASSO SCIENCE INSTITUTE, L'AQUILA



UNIVERSITAT DE
BARCELONA



Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA



EXCELENCIA
MARÍA
DE MAEZTU
2020 - 2024

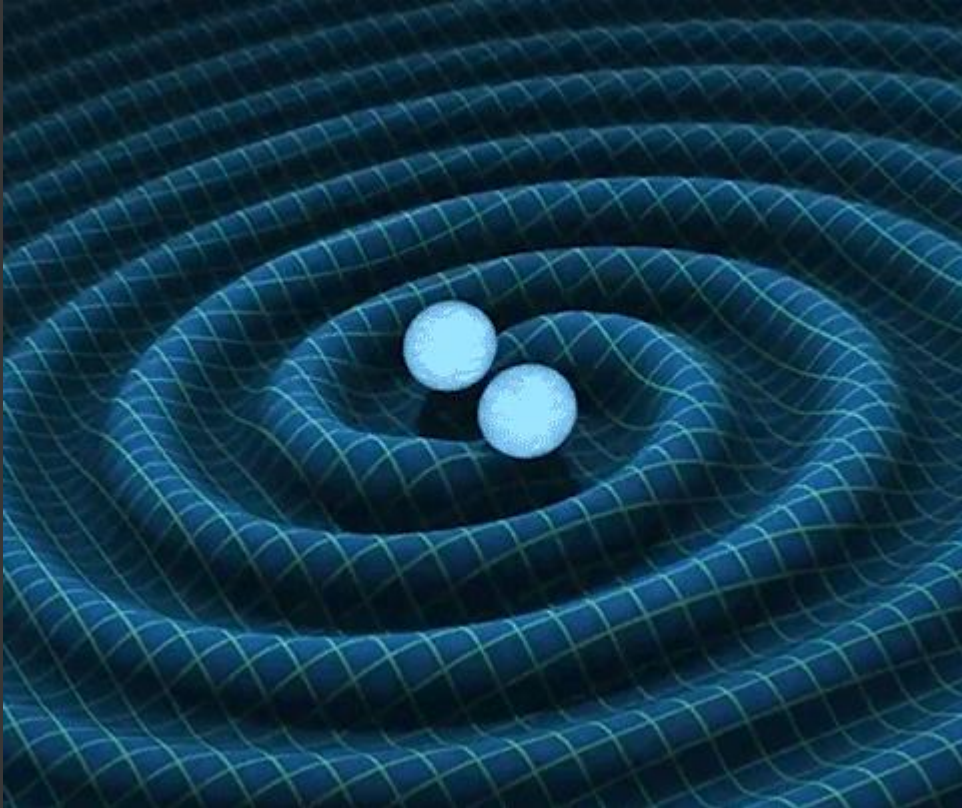


MINISTERIO
DE CIENCIA, INNOVACIÓN
Y UNIVERSIDADES



AGENCIA
ESTATAL DE
INVESTIGACIÓN

NEW WINDOWS TO THE UNIVERSE

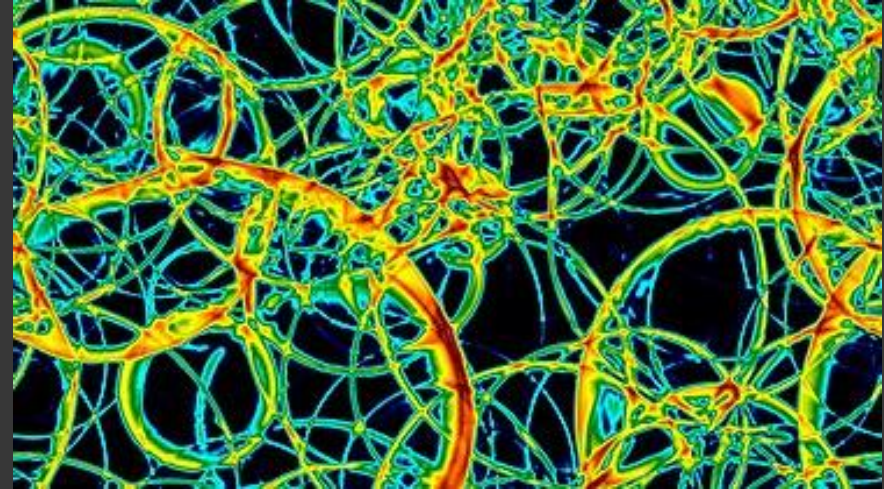


- Strong regime of General Relativity
- Properties of strong-interacting quantum matter

RELEVANT SCENARIOS



Neutron star mergers
SM matter



Phase transition early Universe
BSM matter

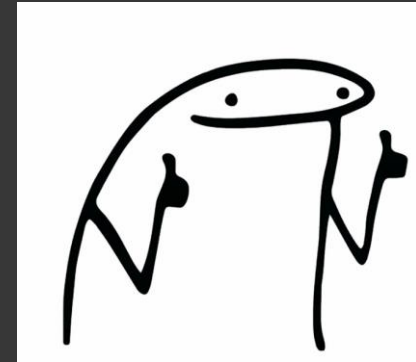
Strongly couple Quantum matter + Out of
Equilibrium



HOLOGRAPHY

WHY HOLOGRAPHY?

- Build the QFT properties from first principles
- Real time dynamics solving Einstein equations in $4+1$ AdS spacetime
- Very difficult to get the $4+1$ AdS spacetime from the QFT properties



Good!



Bad
but...

OUTLINE

1. HOLOGRAPHY IN A NUTSHELL: PRESENTING THE MODEL
2. PHYSICS INFORMED NEURAL NETWORKS (PINNs)
3. RESULTS
4. CONCLUSIONS

THE PEOPLE

BASED ON:
GRAVITATIONAL DUALS FROM EQUATIONS OF STATE
2403.14763 [HEP-TH], 10.1007/JHEP07(2024)087



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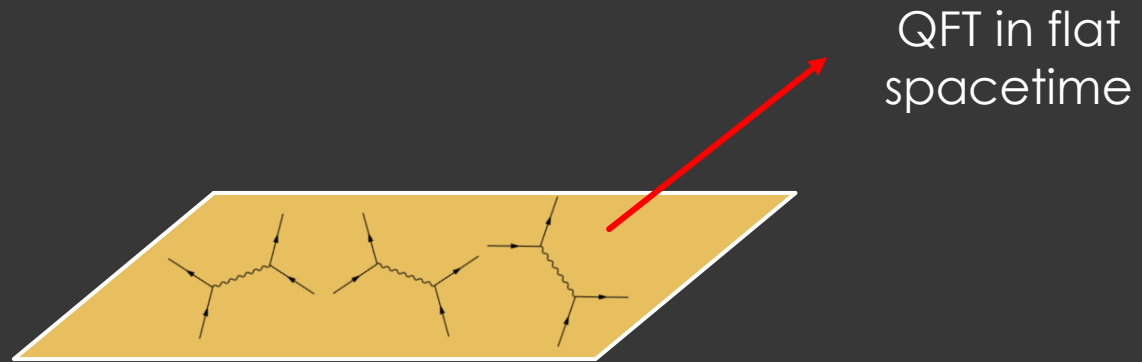


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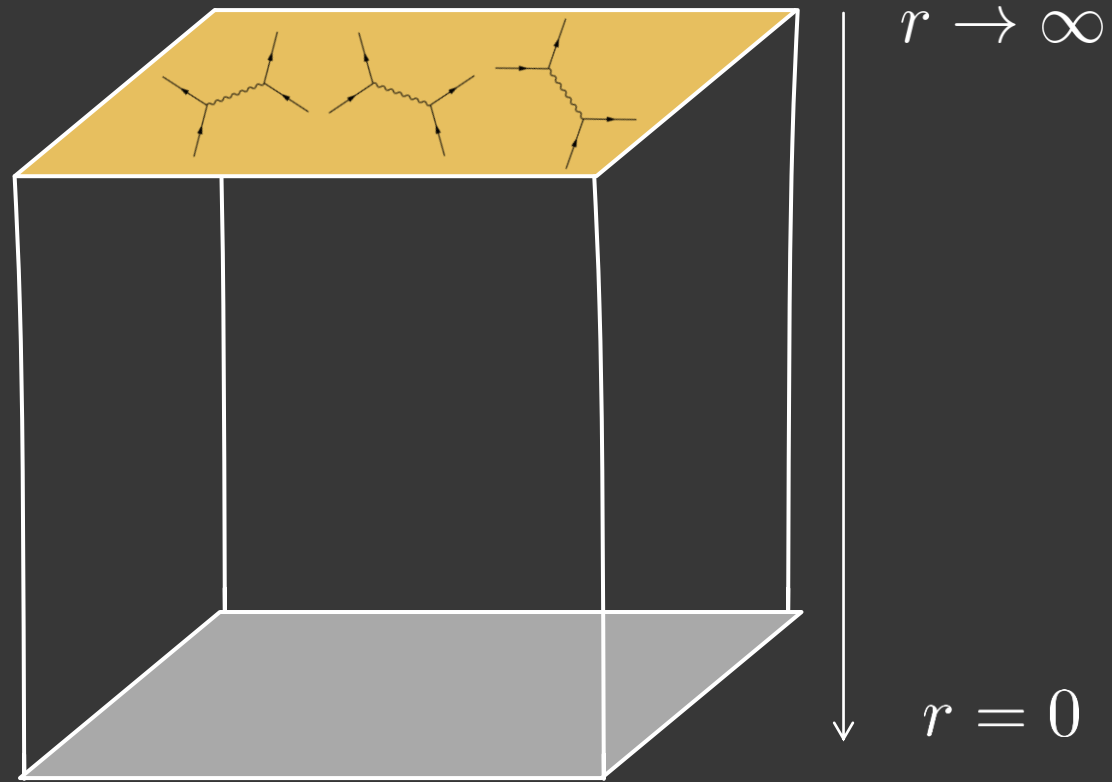


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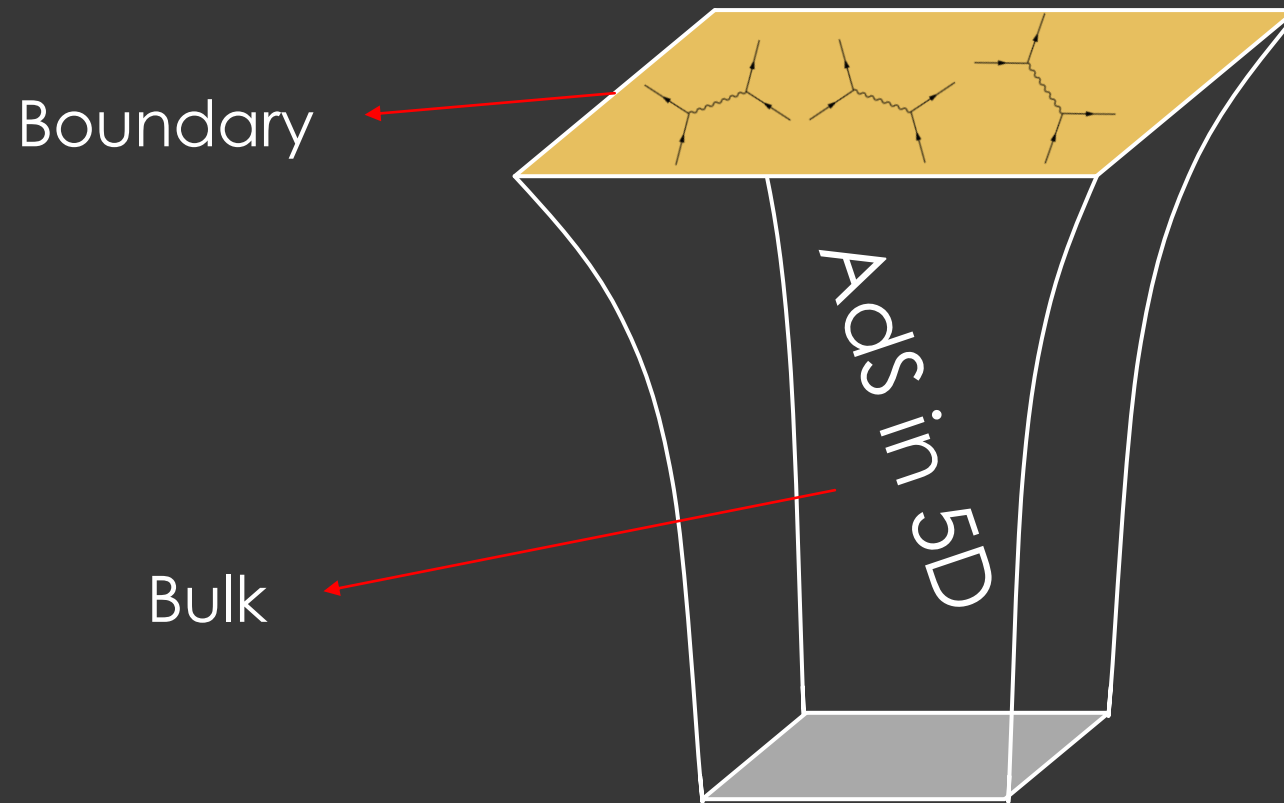
HOLOGRAPHY IN A NUTSHELL



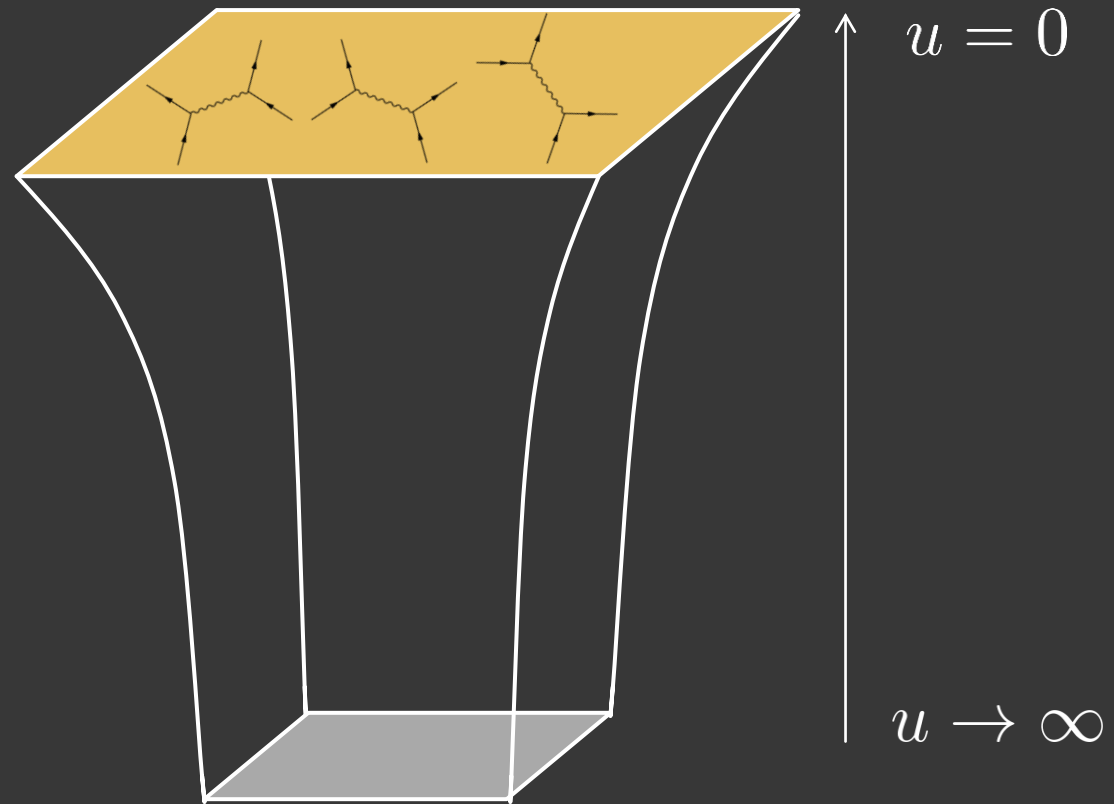
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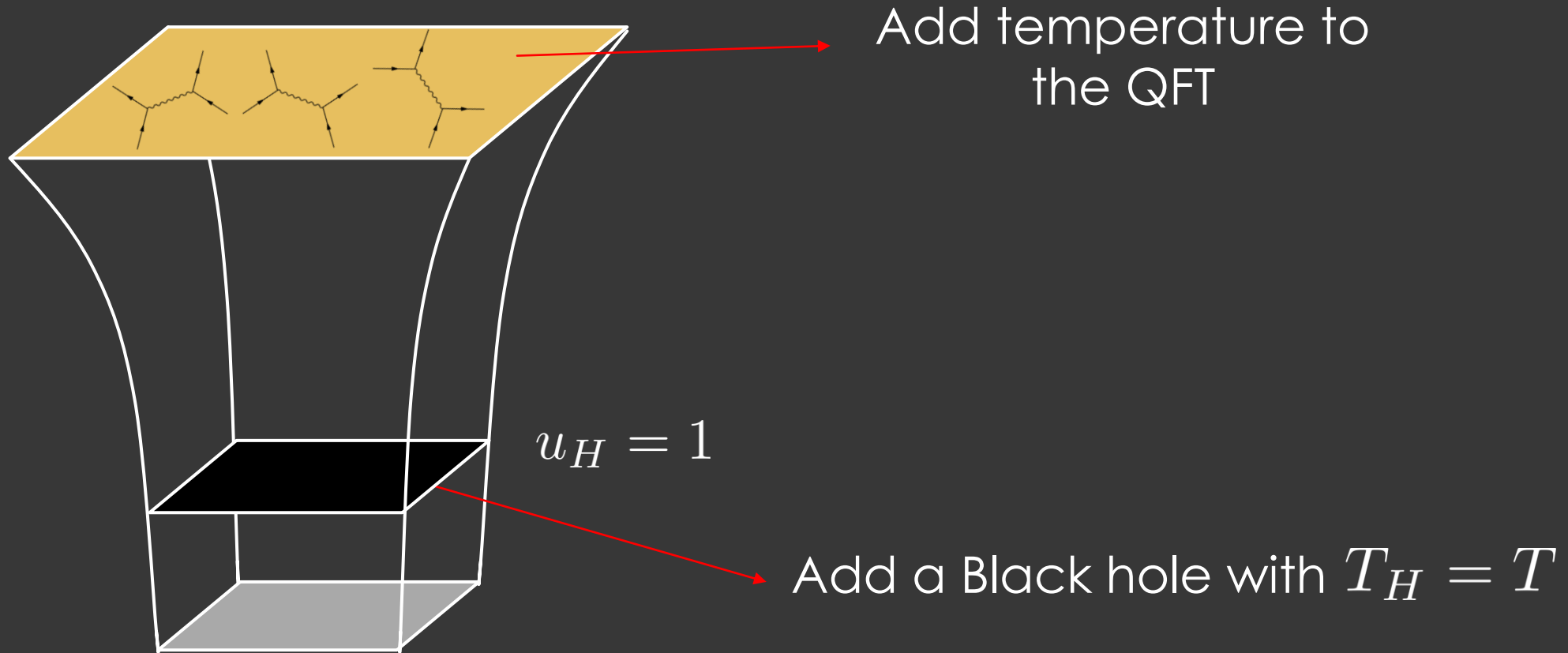
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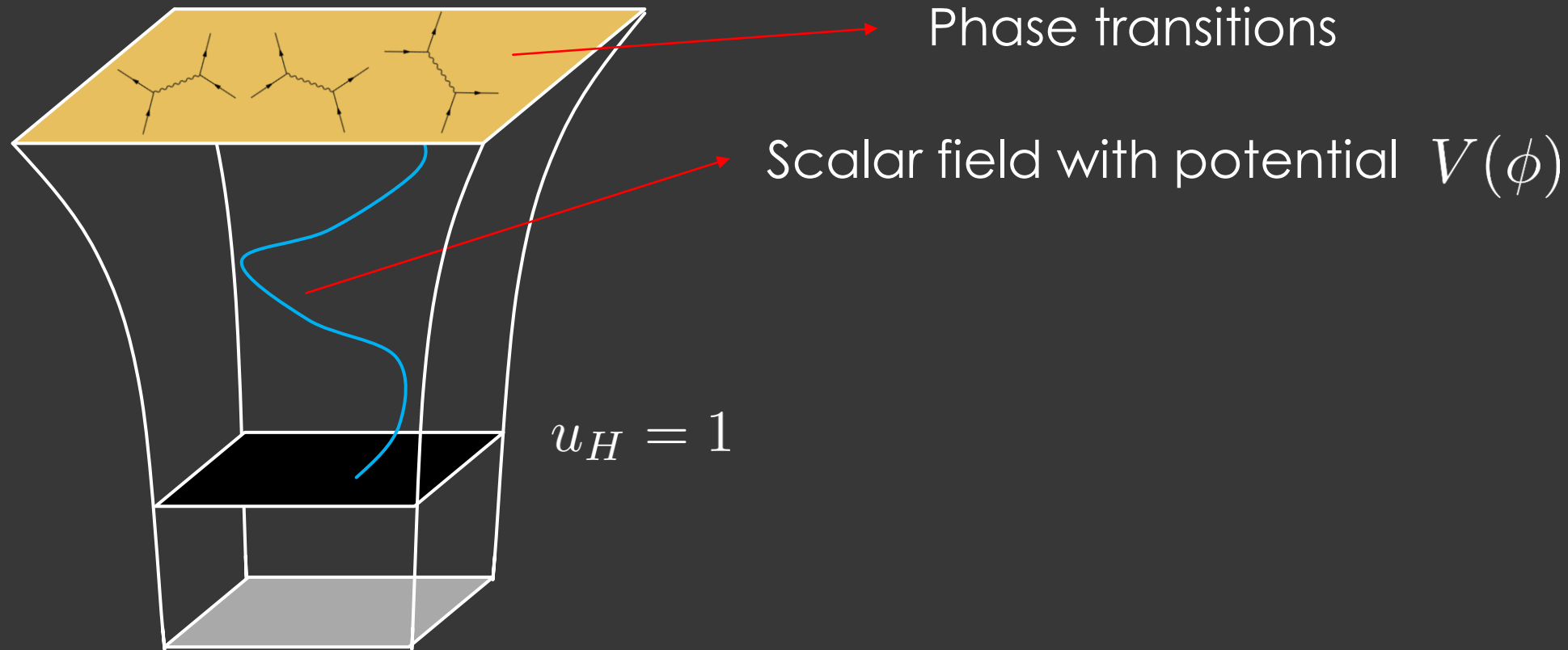
HOLOGRAPHY IN A NUTSHELL



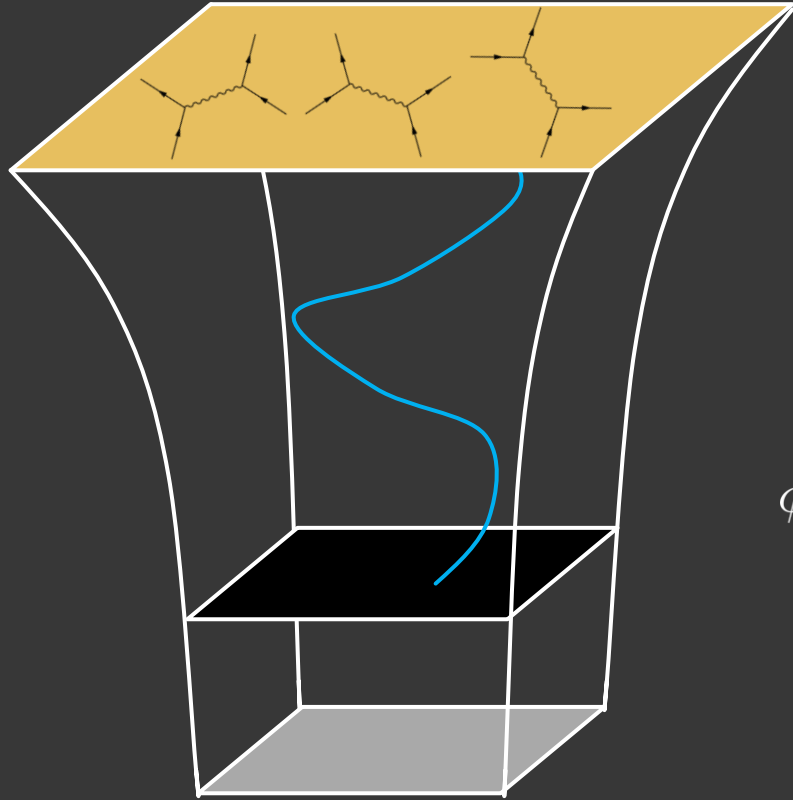
HOLOGRAPHY IN A NUTSHELL



HOLOGRAPHY IN A NUTSHELL



HOLOGRAPHY IN A NUTSHELL



$$ds^2 = -Adt^2 + \Sigma^2(dx^2 + dy^2 + dz^2) - \frac{2}{u^2} dt du$$

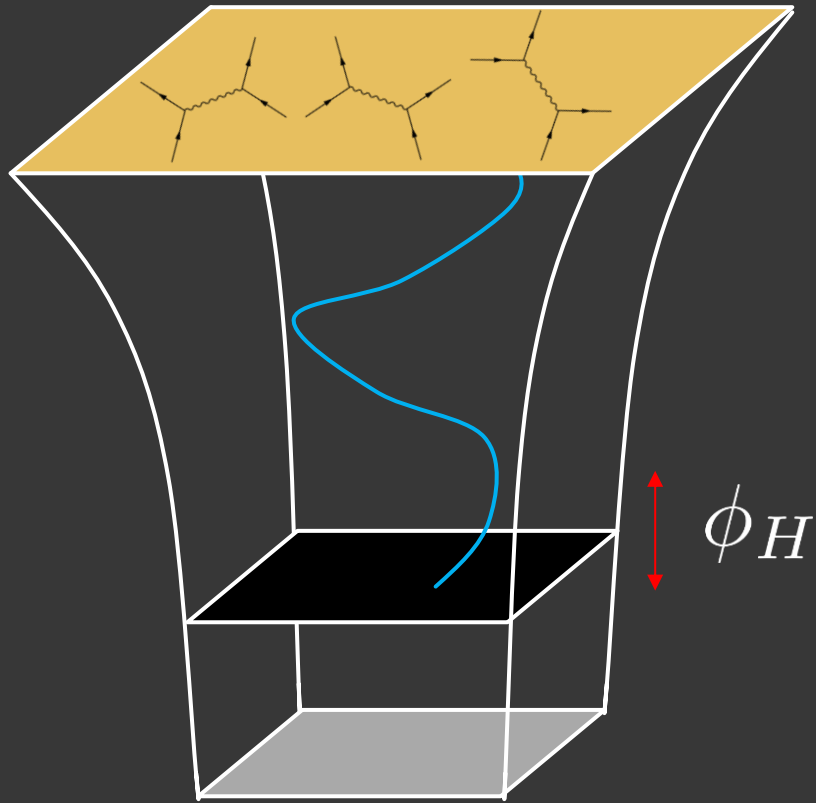
$$\Sigma'' + \frac{2}{u}\Sigma' + \frac{2}{3}\Sigma\phi'^2 = 0 ,$$

$$A'' + \frac{8}{3u^4}V(\phi) + A' \left(\frac{2}{u} + \frac{3\Sigma'}{\Sigma} \right) = 0 ,$$

$$\phi'' - \frac{1}{u^4 A} \frac{\partial V(\phi)}{\partial \phi} + \phi' \left(\frac{2}{u} + \frac{A'}{A} + 3\frac{\Sigma'}{\Sigma} \right) = 0 ,$$

$$A' + 2A\frac{\Sigma'}{\Sigma} - \frac{2\Sigma}{3\Sigma'} \left(A\phi'^2 - \frac{2}{u^4}V(\phi) \right) = 0 ,$$

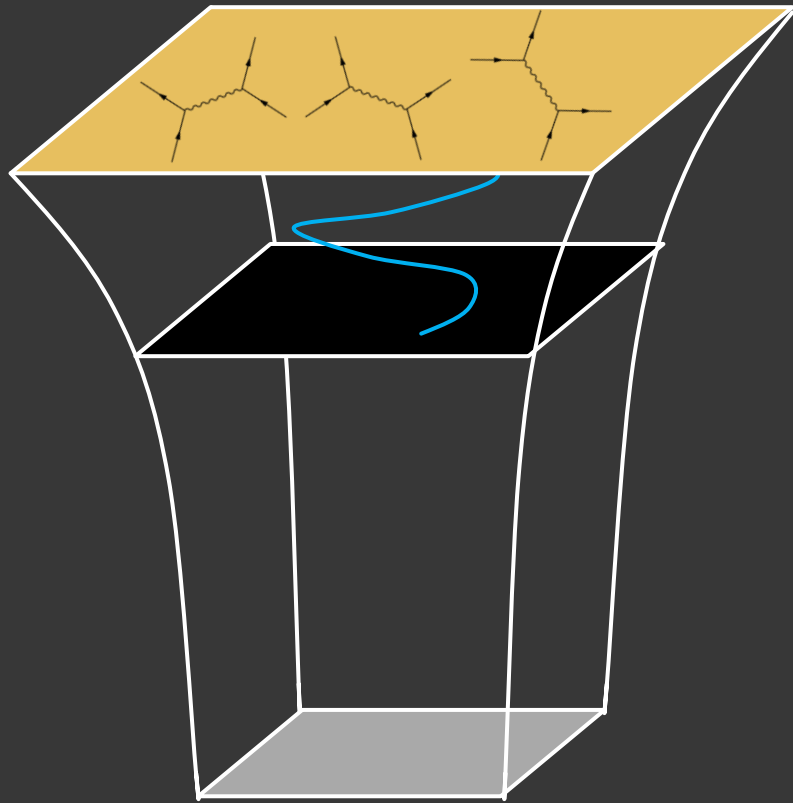
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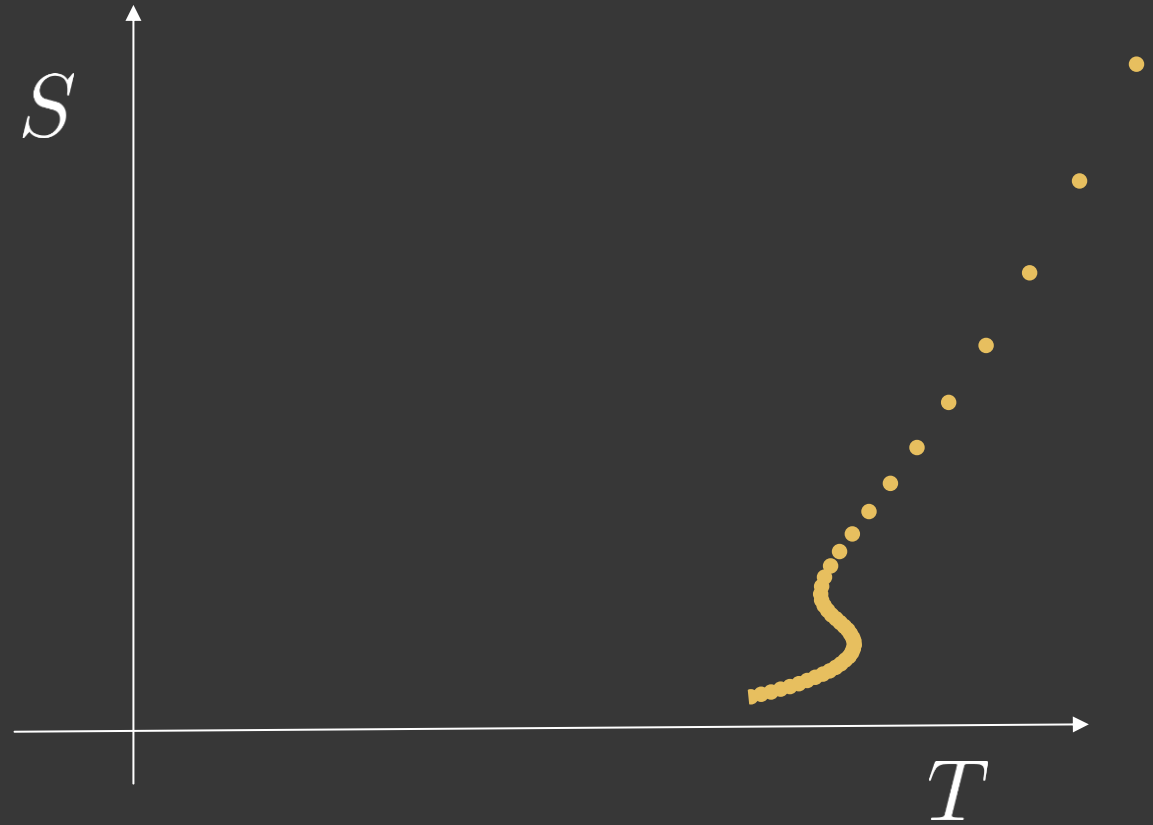
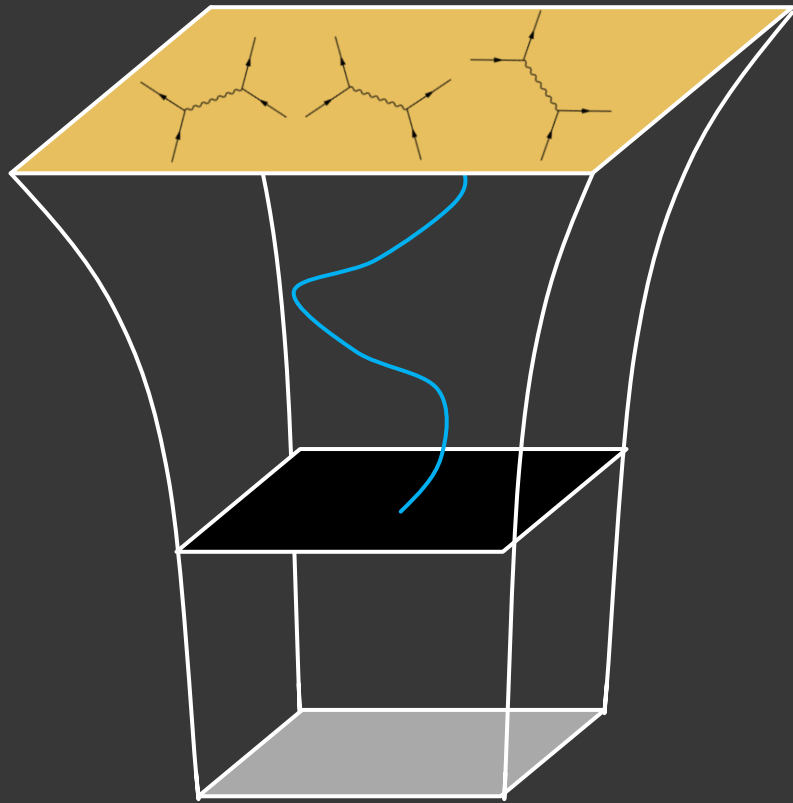
1. Construct different black brane solutions
2. Solve the differential equations
3. Read off the thermodynamics as b.c.

$$\frac{T}{\Lambda} = -\frac{1}{4\pi} \frac{A'(u_H)}{\Lambda}, \quad \frac{S}{\Lambda^3} = \frac{2\pi}{\kappa^2} \left(\frac{\Sigma(u_H)}{\Lambda} \right)^3$$

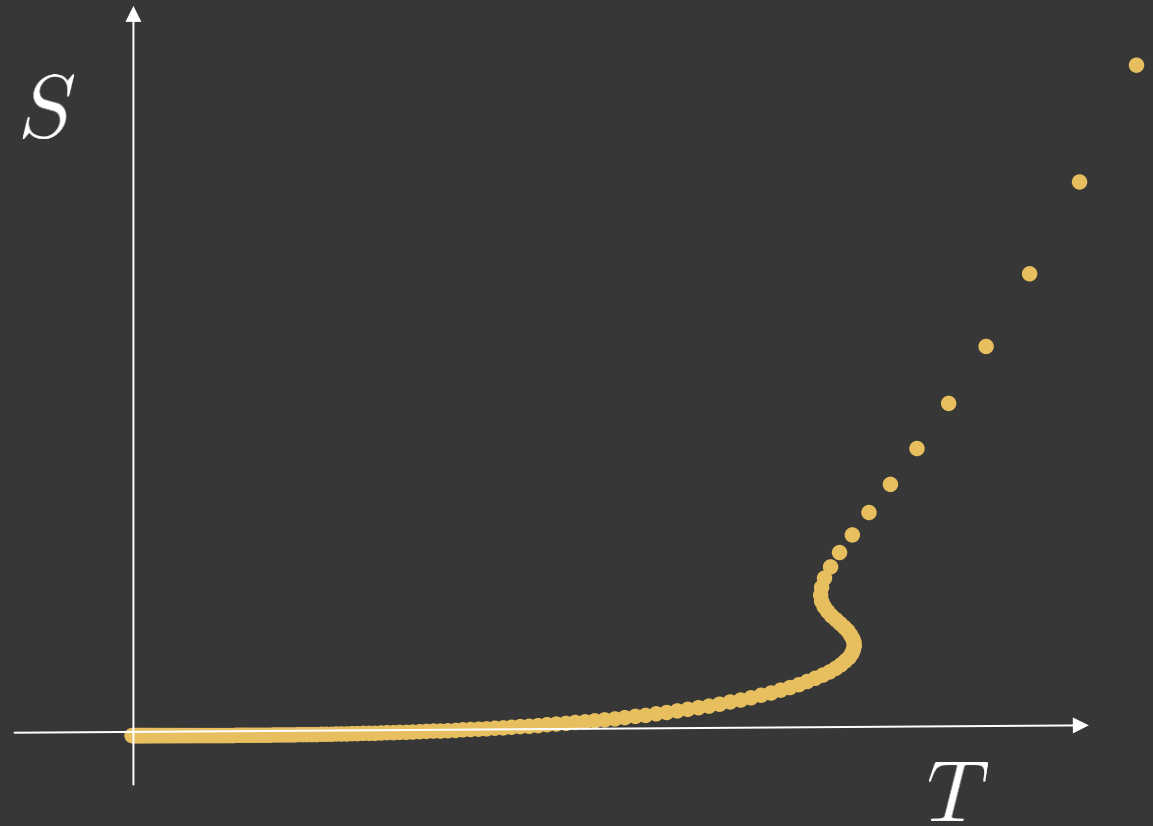
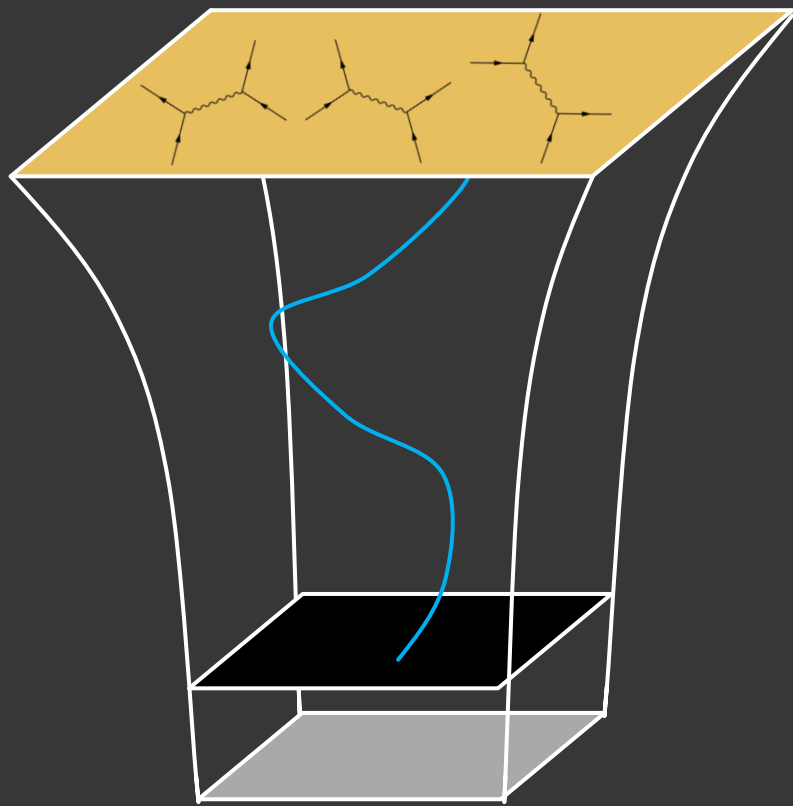
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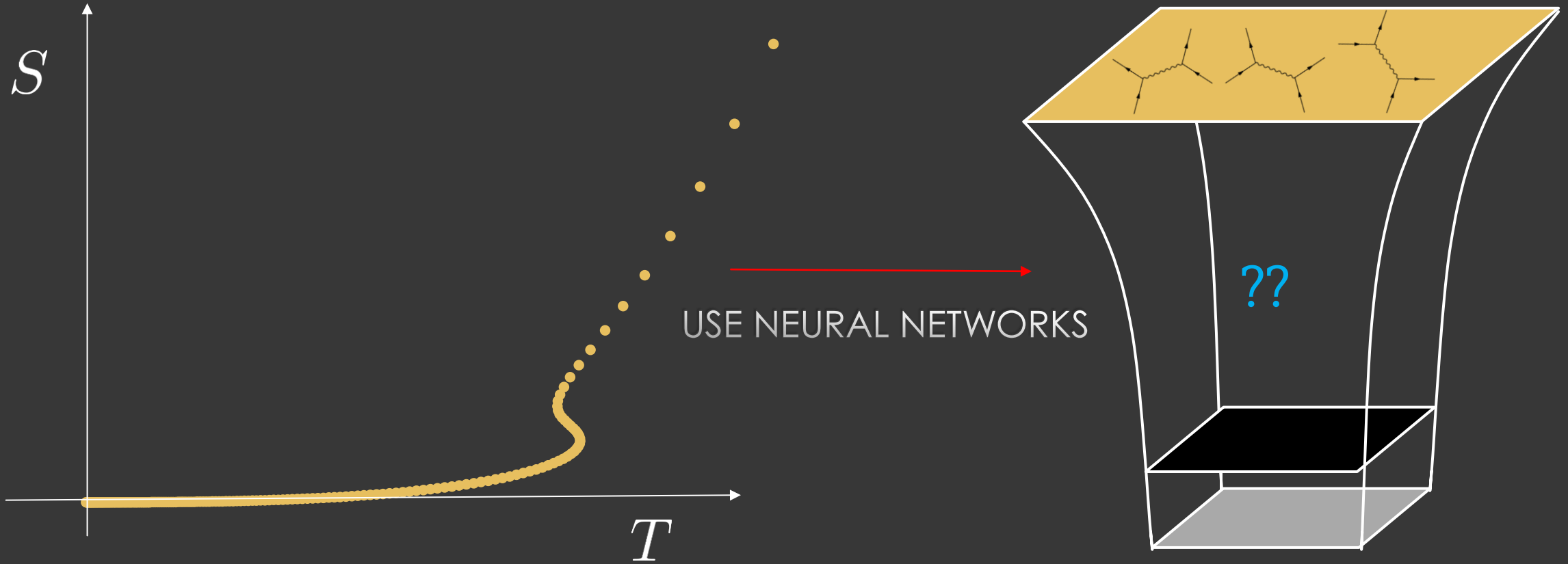
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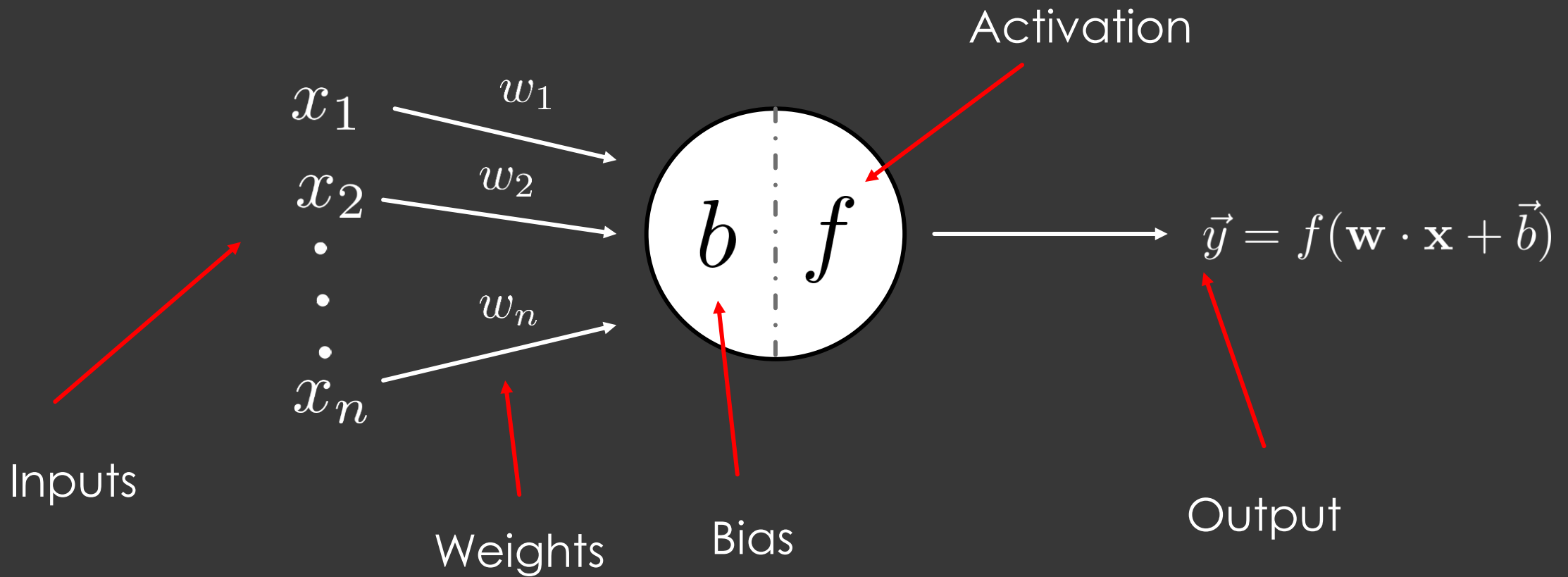
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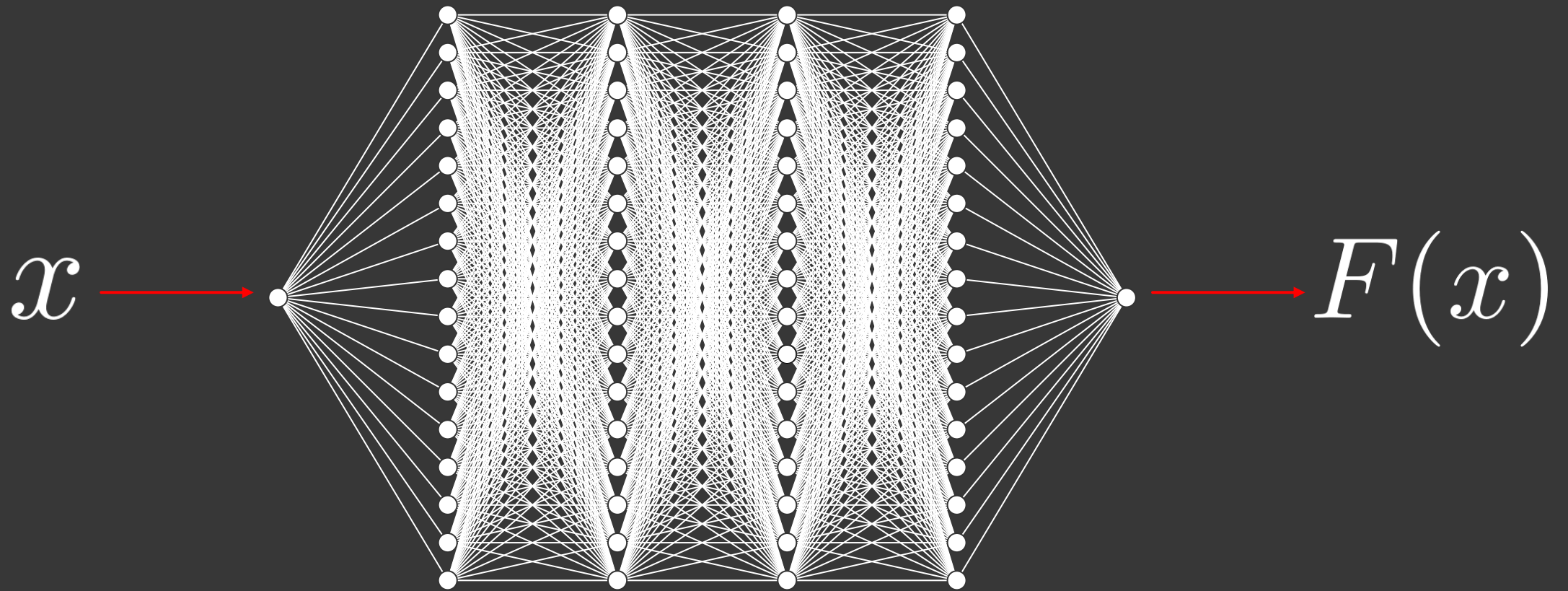
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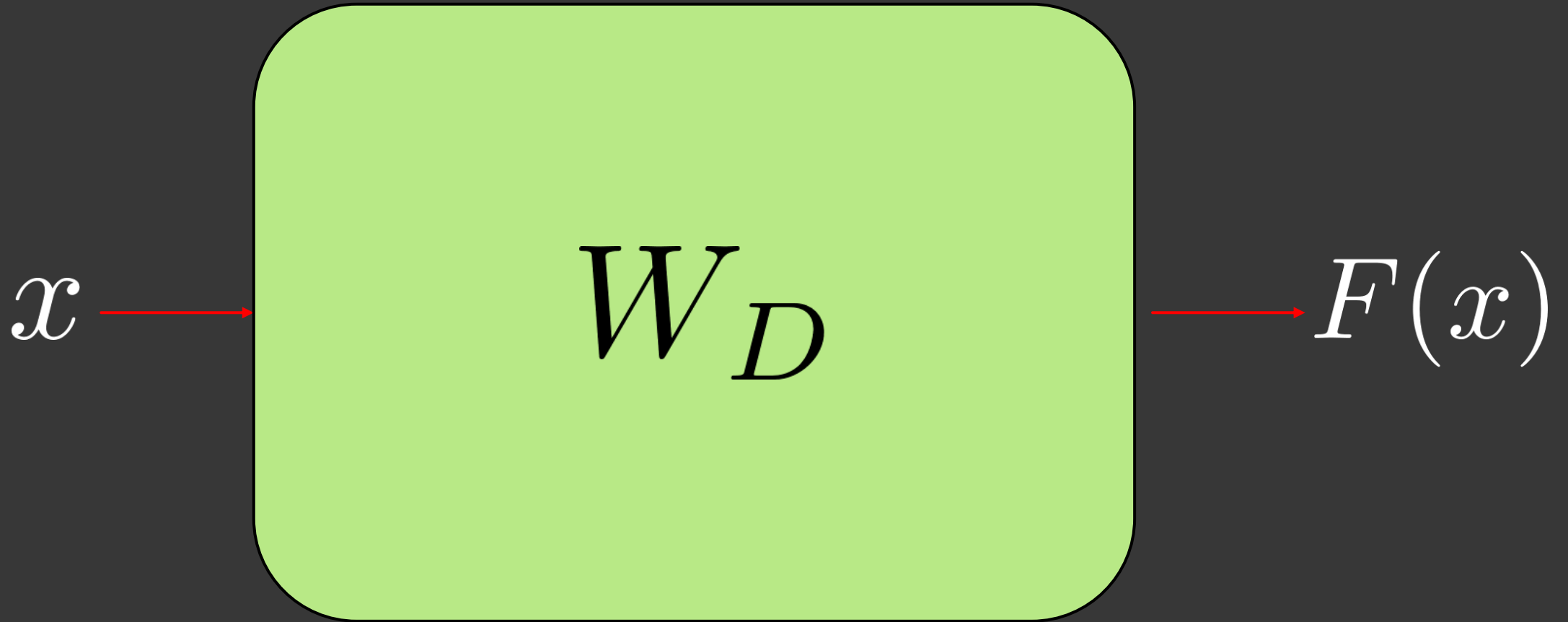
PHYSICS INFORMED NEURAL NETWORKS



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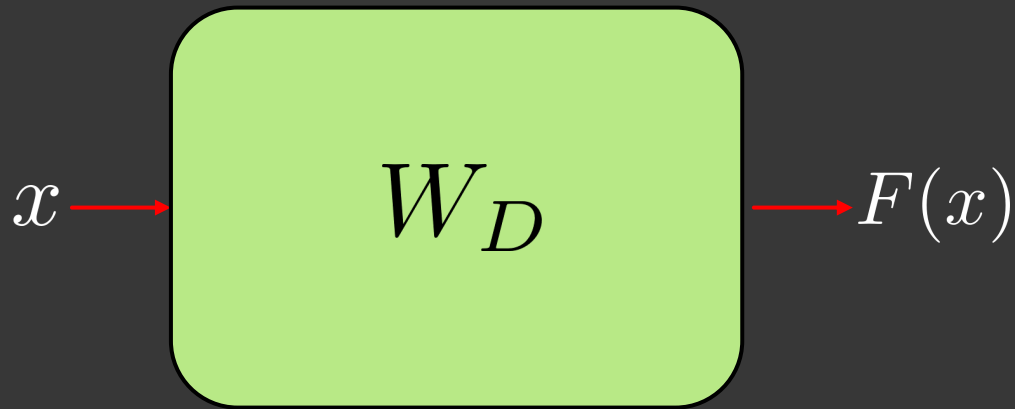
PHYSICS INFORMED NEURAL NETWORKS



PHYSICS INFORMED NEURAL NETWORKS

[Dissanayake, M and
Phan-Thien, N. 1994]

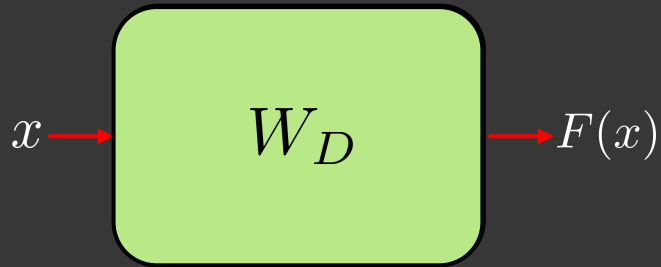
[Lagaris et al. 1997]



[Chen et al. 2020]

- Use a NN for each unknown function
- B.C are fulfilled exactly by construction
- Derivatives are computed analytically
- Slow training but evaluation is instantaneous
- Perfectly suited for inverse problems
- Train for different boundary conditions at once (solution bundle)
- Neurodiffeq

PHYSICS INFORMED NEURAL NETWORKS



Equations

$$E_1 = \nu_\Sigma - \tilde{\Sigma}',$$

$$E_2 = \nu_A - \tilde{A}',$$

$$E_3 = \nu_\phi - \phi',$$

$$E_4 = \nu'_\Sigma + \frac{2}{3}\tilde{\Sigma}\nu_\phi^2,$$

$$E_5 = u^2 \tilde{\Sigma} \nu'_A + \frac{8}{3} V(\phi) \tilde{\Sigma} + \nu_A \left(3u^2 \nu_\Sigma - 5u \tilde{\Sigma} \right) + \tilde{A} \left(8\tilde{\Sigma} - 6u \nu_\Sigma \right),$$

$$E_6 = u^2 \tilde{\Sigma} \tilde{A} \nu'_\phi - \tilde{\Sigma} \frac{dV}{d\phi} + \nu_\phi \left(-3u \tilde{A} \tilde{\Sigma} + u^2 \tilde{\Sigma} \nu_A + 3u^2 \nu_\Sigma \tilde{A} \right),$$

$$E_7 = \left(u \nu_\Sigma - \tilde{\Sigma} \right) \left(u^2 \tilde{\Sigma} \nu_A + 2u^2 \tilde{A} \nu_\Sigma - 4u \tilde{A} \tilde{\Sigma} \right) - \frac{2}{3} u \tilde{\Sigma}^2 \left(u^2 \tilde{A} \nu_\phi^2 - 2V(\phi) \right)$$

Loss function

$$L \equiv \sum_{\alpha} \sum_n \sum_i E_{\alpha} \left(u_n, (T_i, S_i) \right)^2 + V_{BC}.$$

Boundary conditions

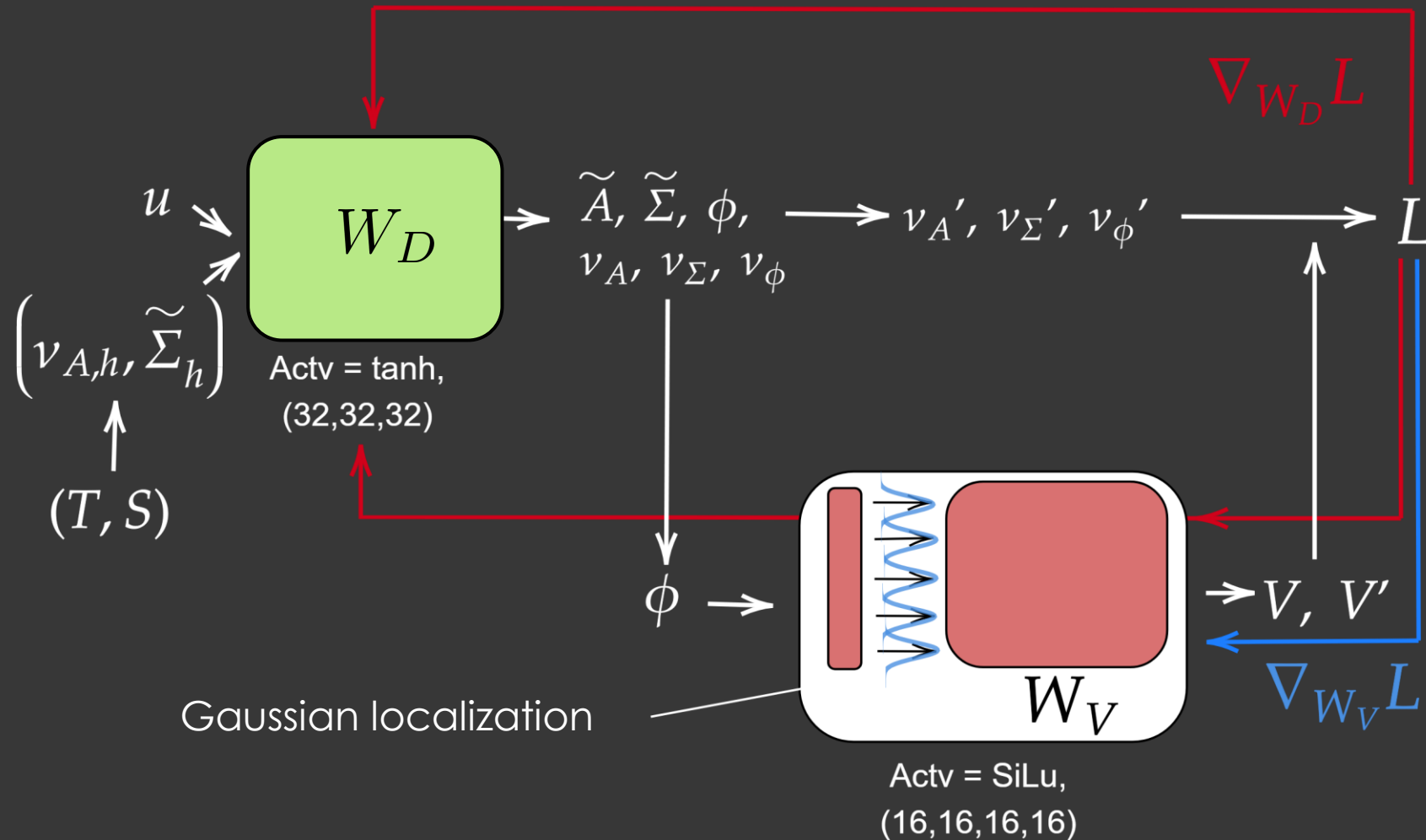
$$\tilde{A}|_{u=0} = 1, \quad \tilde{A}|_{u=1} = 0,$$

$$\tilde{\Sigma}|_{u=0} = 1, \quad \nu_A|_{u=1} = -4\pi T,$$

$$\phi|_{u=0} = 0, \quad \tilde{\Sigma}_{u=1} = (S/\pi)^{1/3}.$$

$$\nu_\phi|_{u=0} = 1,$$

PHYSICS INFORMED NEURAL NETWORKS



PHYSICS INFORMED NEURAL NETWORKS

Gaussian localization

$$\mathbf{h}_i = \mathbf{h}_{i-1} \cdot \mathbf{W}_{\mathbf{V},i}^T$$

$$\tilde{\mathbf{h}}_1 = \text{SiLU} \left\{ \mathbf{h}_1 \times \exp \left[- \left(\frac{\mathbf{h}_1 - \mu}{\sigma} \right)^2 \right] \right\}$$

PHYSICS INFORMED NEURAL NETWORKS

Gaussian localization

$$\mathbf{h}_i = \mathbf{h}_{i-1} \cdot \mathbf{W}_{\mathbf{V},i}^T$$

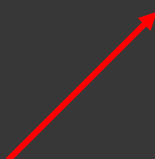
$$\tilde{\mathbf{h}}_1 = \text{SiLU} \left\{ \mathbf{h}_1 \times \exp \left[- \left(\frac{\mathbf{h}_1 - \mu}{\sigma} \right)^2 \right] \right\}$$

- Fixed sigma's but learnable mu's
- Each neuron of the first layer will specialize on a certain region of the input domain
- Separate loss gradient for inputs that are far apart
- Multi-scale $\rightarrow$ large curvature differences between the boundary and near the horizon region.
- Multi-entangled $\rightarrow$ low-temperature region includes information about the high temperature region

RESULTS

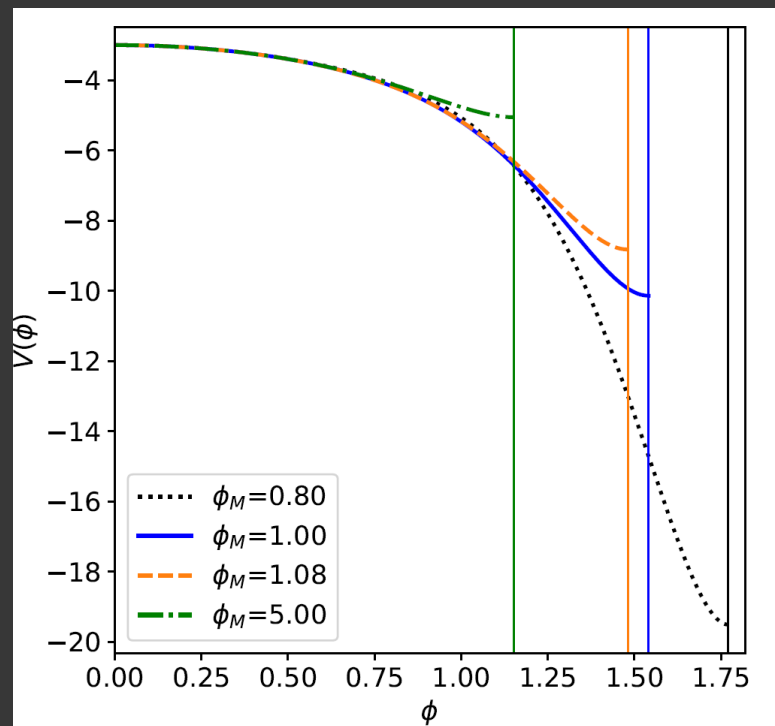
Family of potentials with a free parameter ϕ_M

$$V_{\text{theory}}(\phi) = -\frac{4}{3}\mathcal{W}(\phi)^2 + \frac{1}{2}\mathcal{W}'(\phi)^2$$

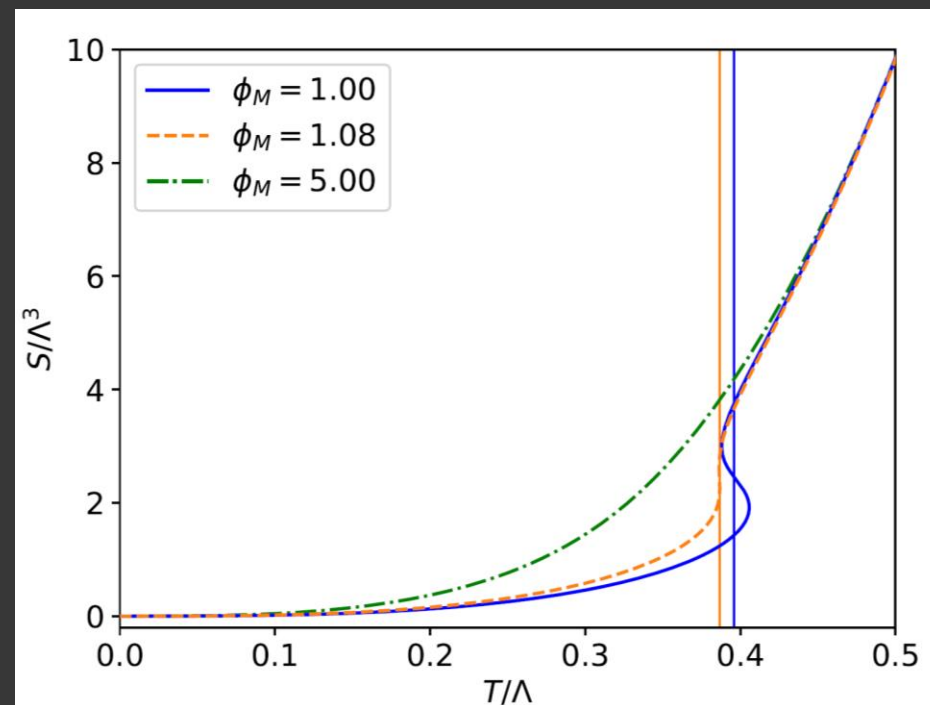
$$\mathcal{W}(\phi) = -\frac{3}{2} - \frac{\phi^2}{2} - \frac{\phi^4}{4\phi_M^2} + \frac{\phi^6}{10}$$


Regulates the phase transition in the boundary theory

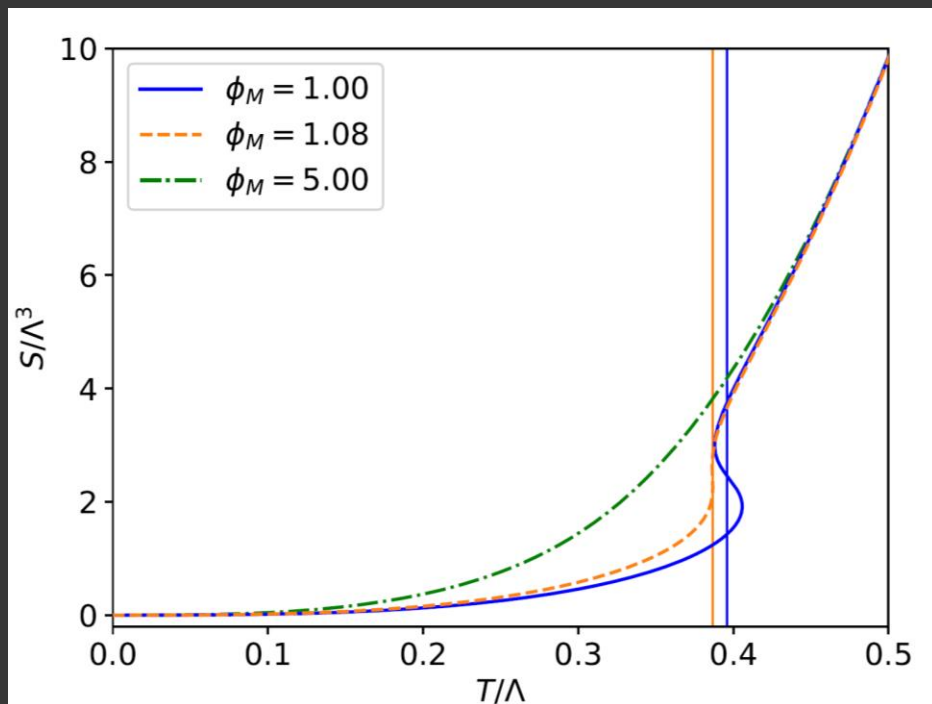
RESULTS



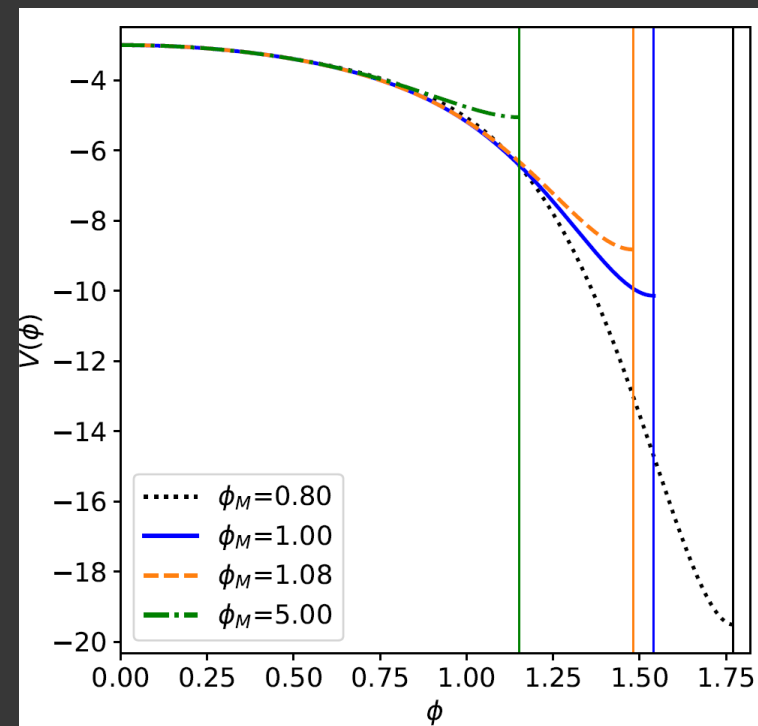
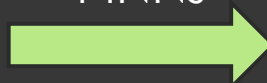
Direct problem



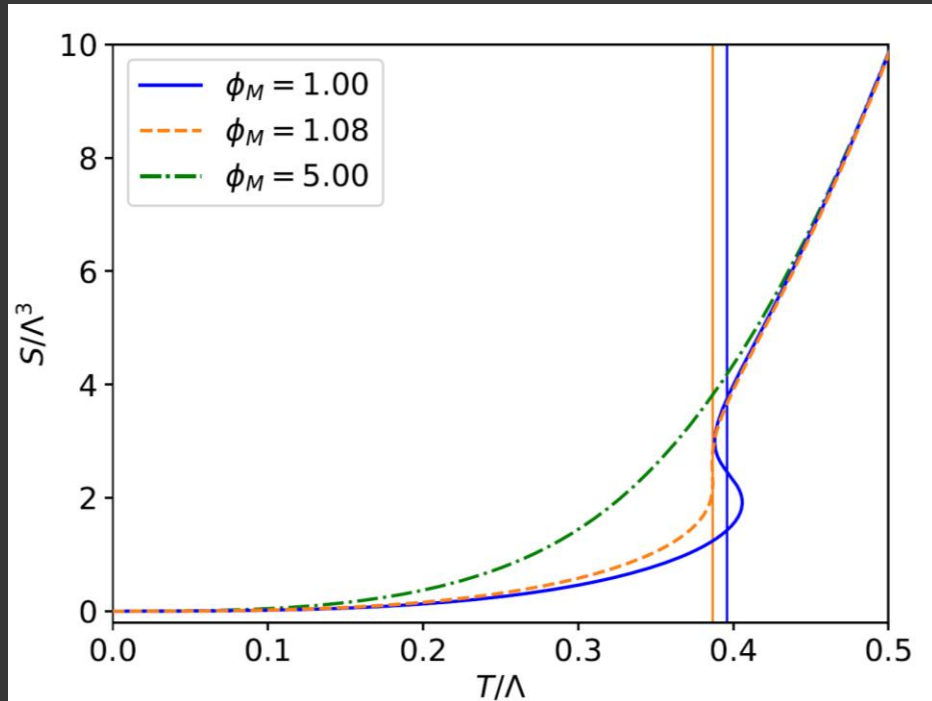
RESULTS



Inverse problem
PINNs

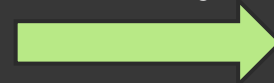


RESULTS

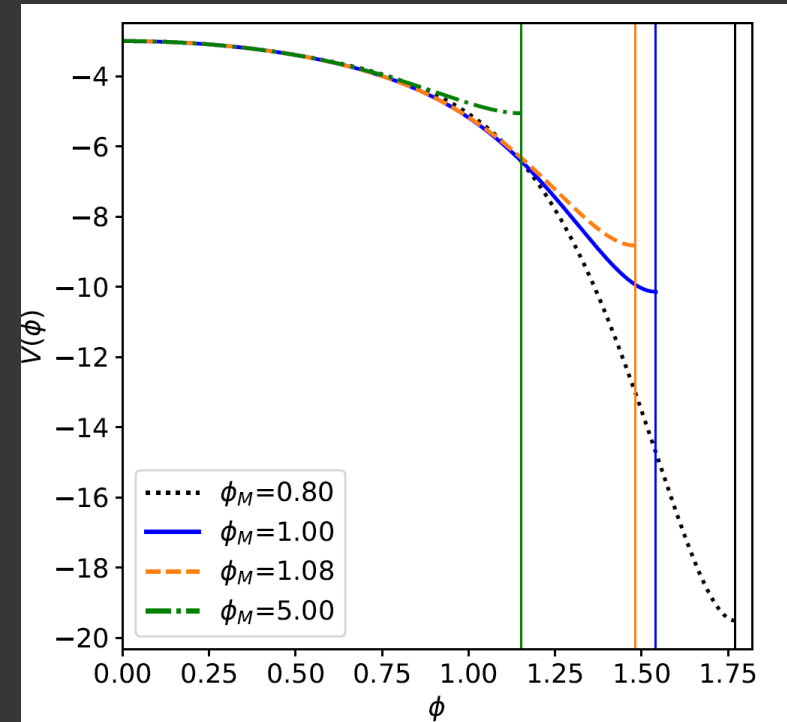


$$|\delta T(S)| = \left| \frac{T_{\text{input}}(S) - T_{\text{PINN}}(S)}{T_{\text{input}}(S)} \right|$$

Inverse problem
PINNs



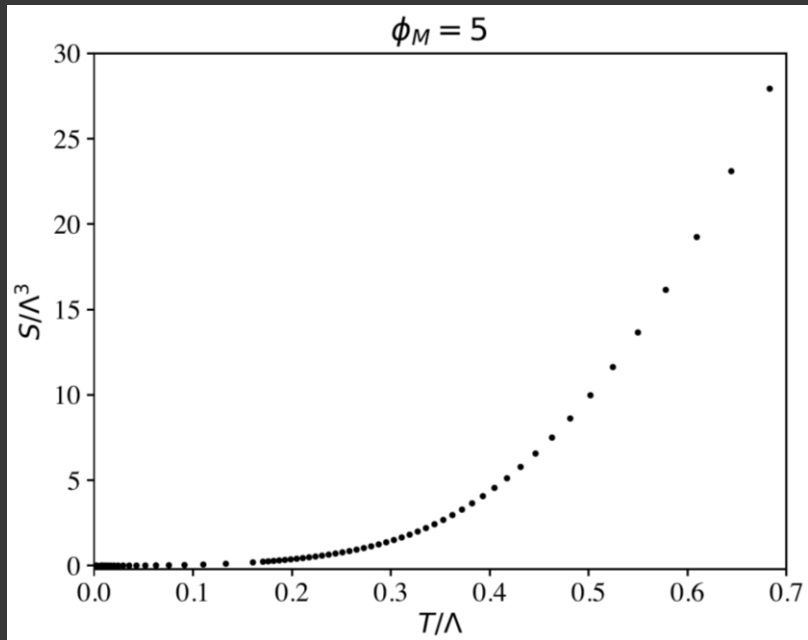
Direct problem
Compare S(T)



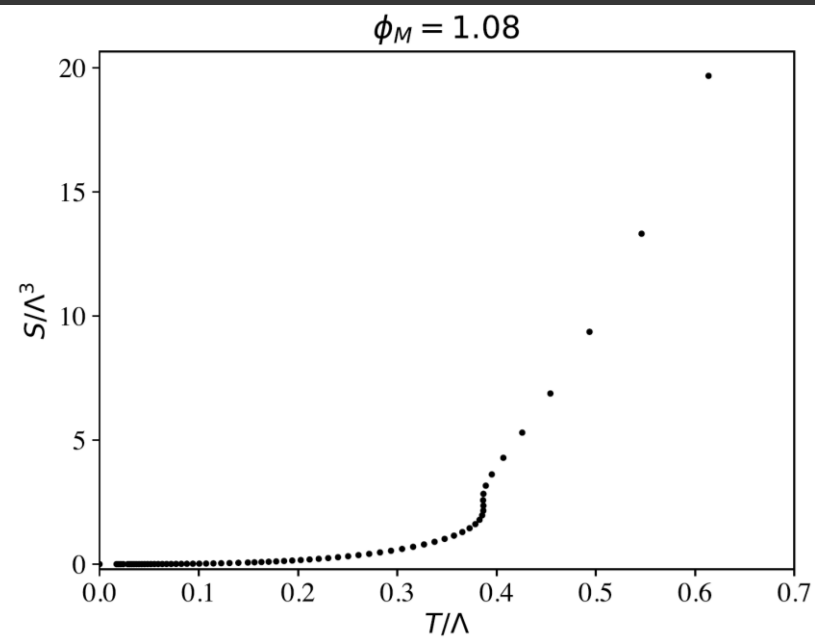
$$|\delta V(\phi)| = \left| \frac{V_{\text{theory}}(\phi) - V_{\text{PINN}}(\phi)}{V_{\text{theory}}(\phi)} \right|$$

RESULTS

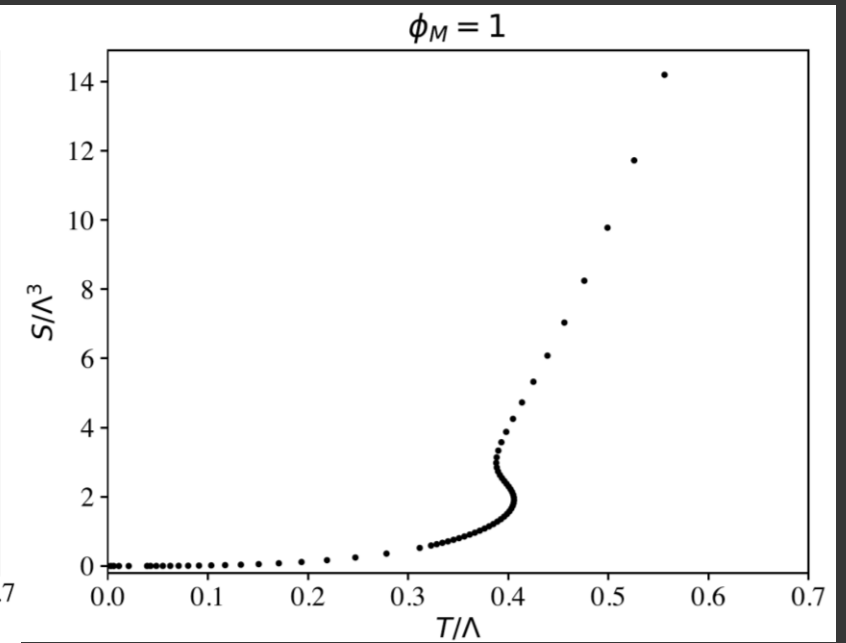
Crossover



Second Order

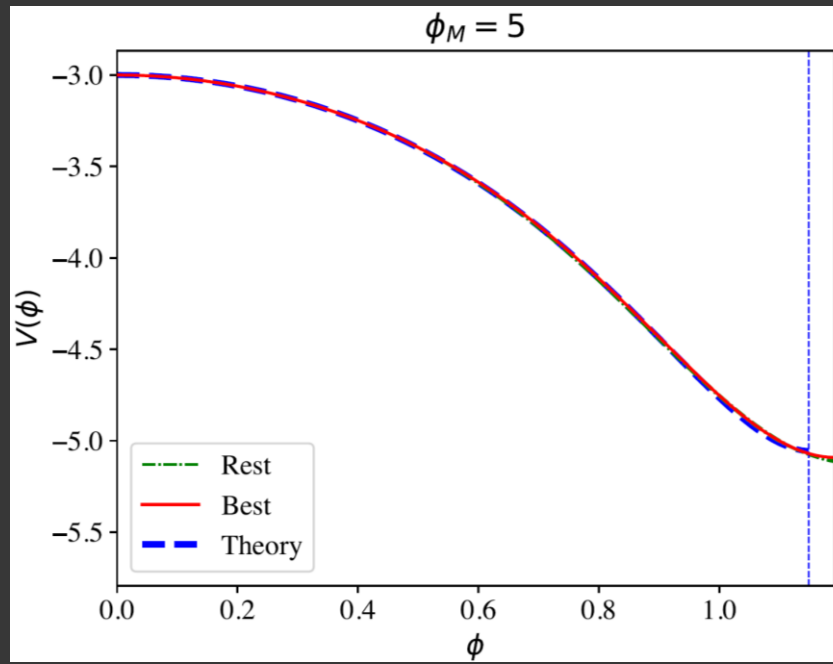


First Order

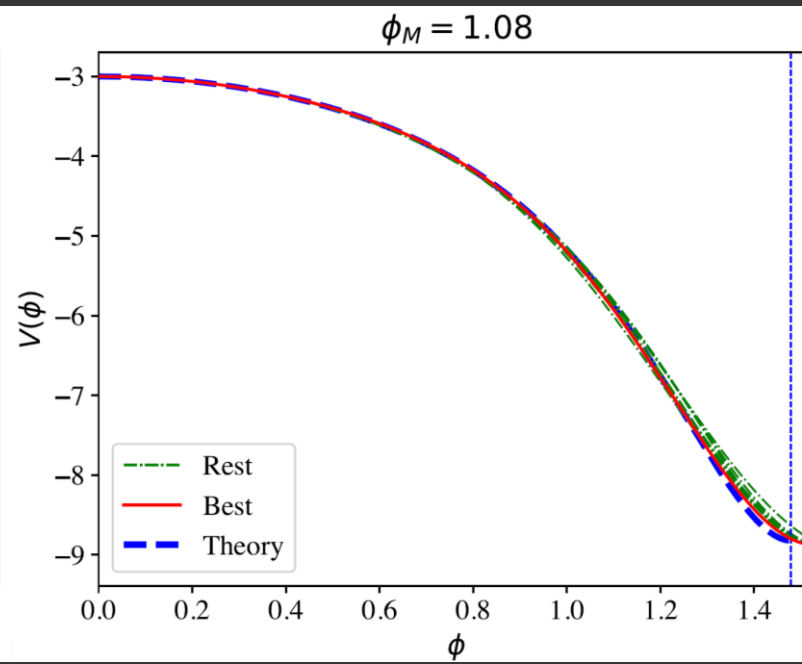


RESULTS

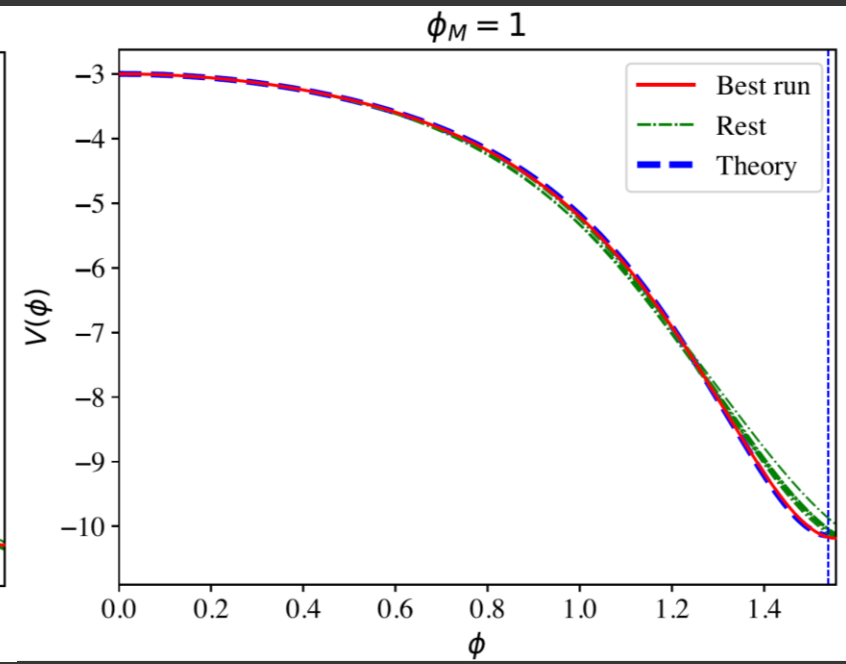
Crossover



Second Order

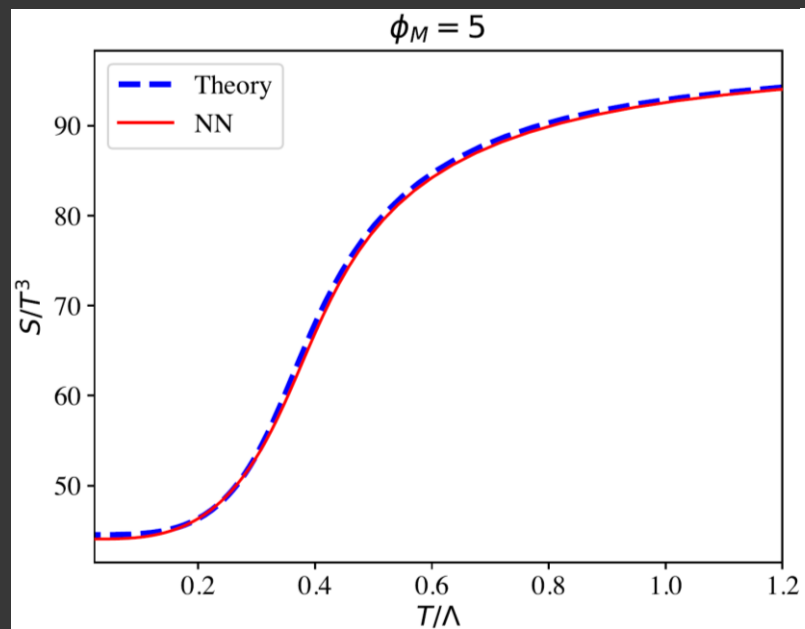


First Order

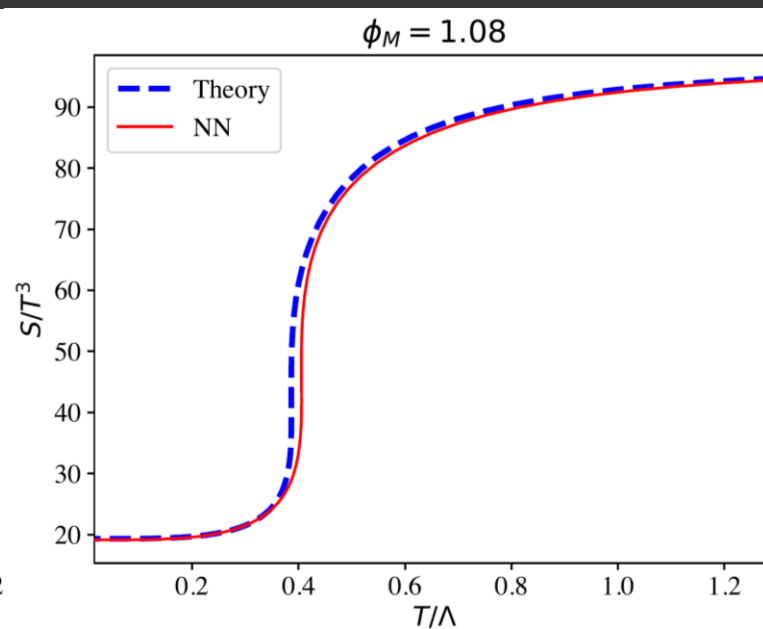


RESULTS

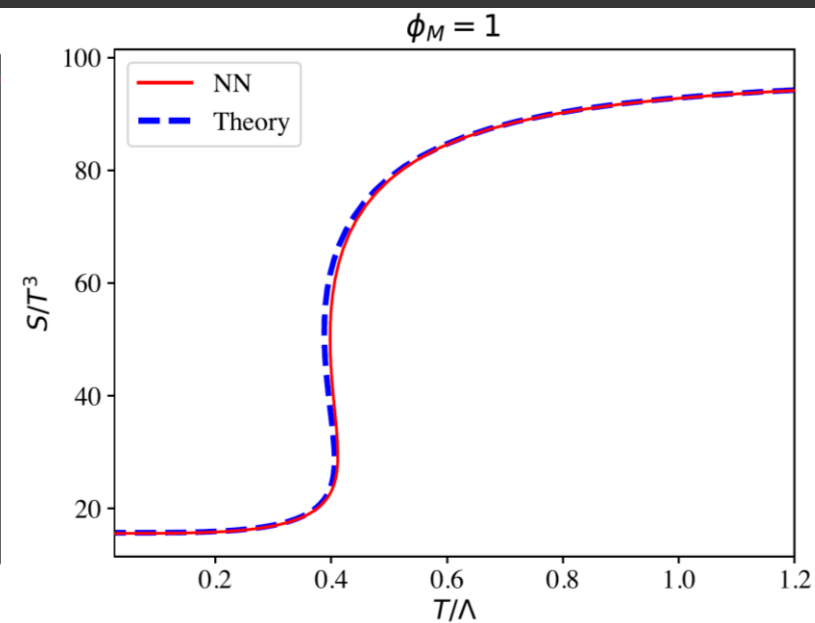
Crossover



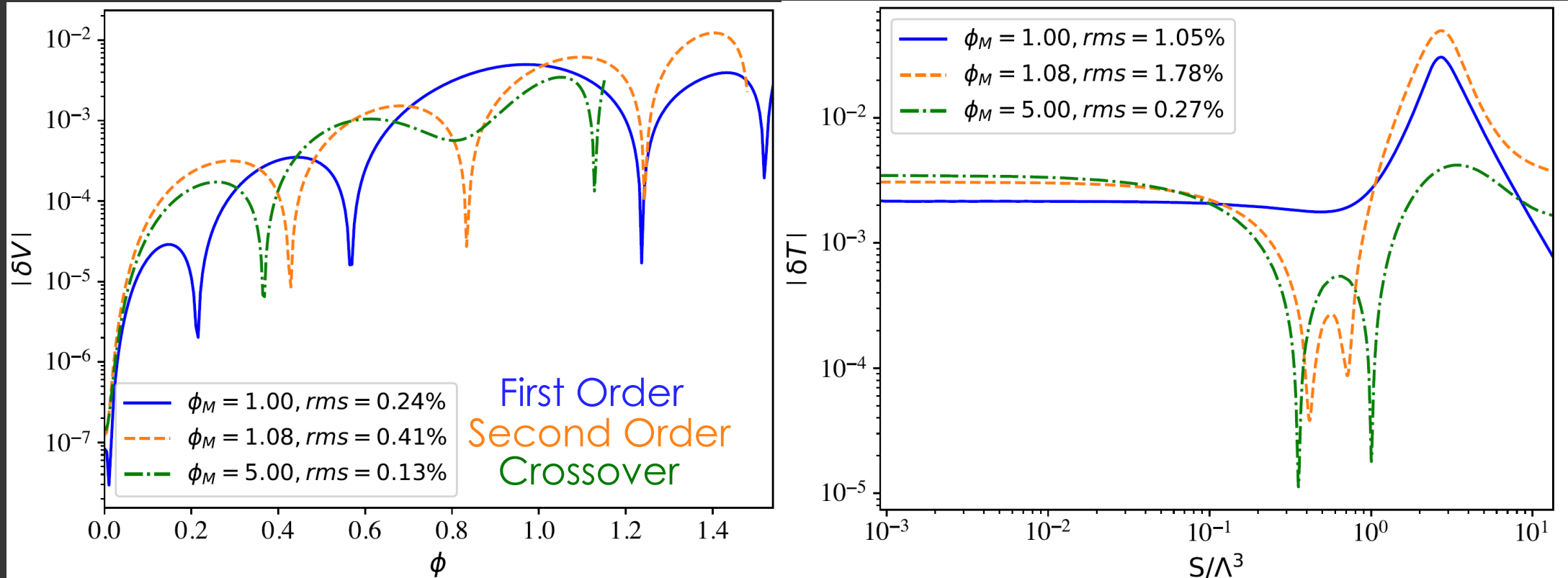
Second Order



First Order



RESULTS



CONCLUSIONS AND FUTURE DIRECTIONS

- It is possible to recover the bulk theory using data from the boundary theory. This is a very non-trivial result in the field of Holography.
- Applicable to many situations: QCD equation of state, physics BSM from GW data, Neutron Star mergers...
- GL helped but it is not enough to solve the stiffest cases: false vacuum decay, naked singularity in the bulk...
- Multihead + flattening the latent space might work for the most difficult cases
- Method generalizable to other inversion problem in physics that involve highly non-linear differential equations, i.e. Navier-Stokes equation.

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DIRECT PROBLEM DETAILS

$$\begin{aligned}A(u) &= \frac{1}{u^2} + \cdots , \\ \Sigma(u) &= \frac{1}{u} + \cdots , \\ \phi(u) &= \Lambda u + \cdots ,\end{aligned}$$

$$\begin{aligned}A(u) &= A_1(u_H - u) + \cdots , \\ \Sigma(u) &= \Sigma_0 + \Sigma_1(u_H - u) + \cdots , \\ \phi(u) &= \phi_H + \phi_1(u_H - u) + \cdots .\end{aligned}$$

POTENTIAL BOUNDARY CONDITIONS

$$S_{\text{theory}}(T) = \pi^4 T^3 + \frac{3\pi^4}{64} T + \dots .$$

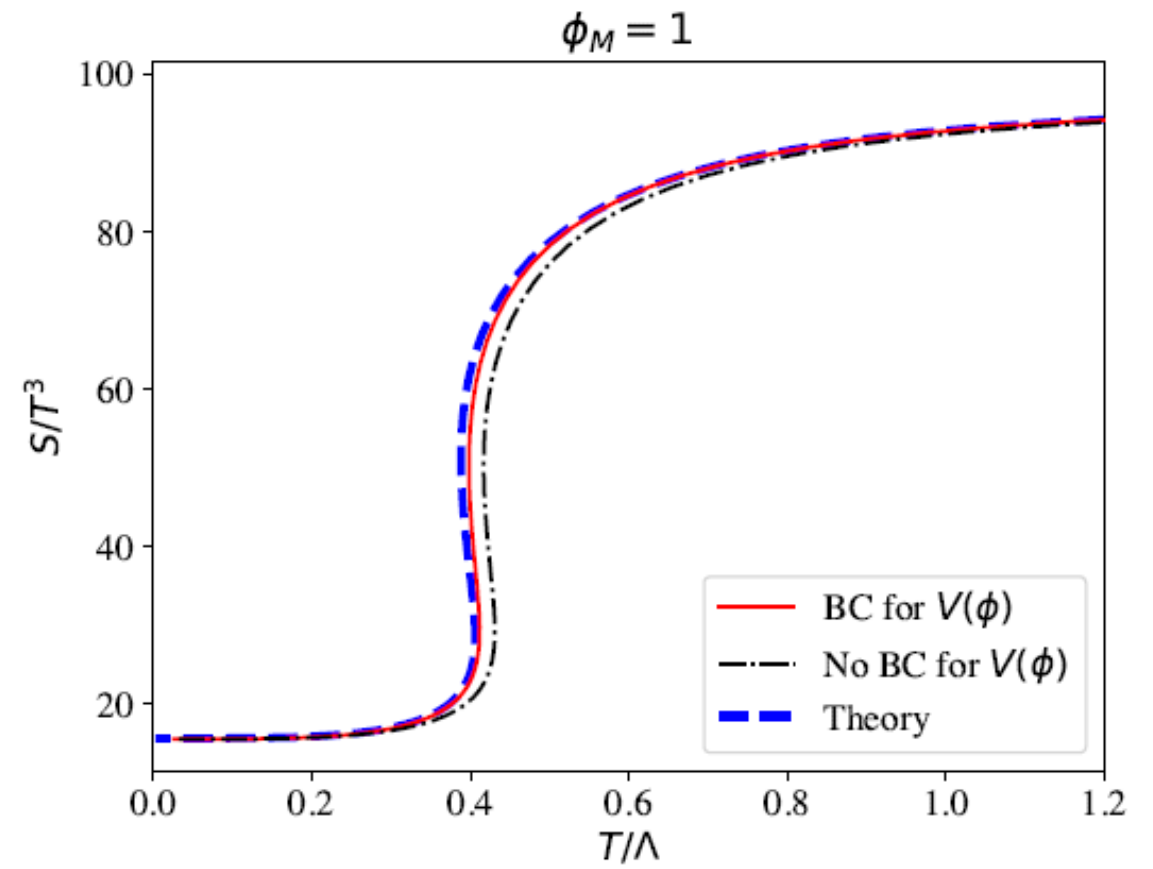
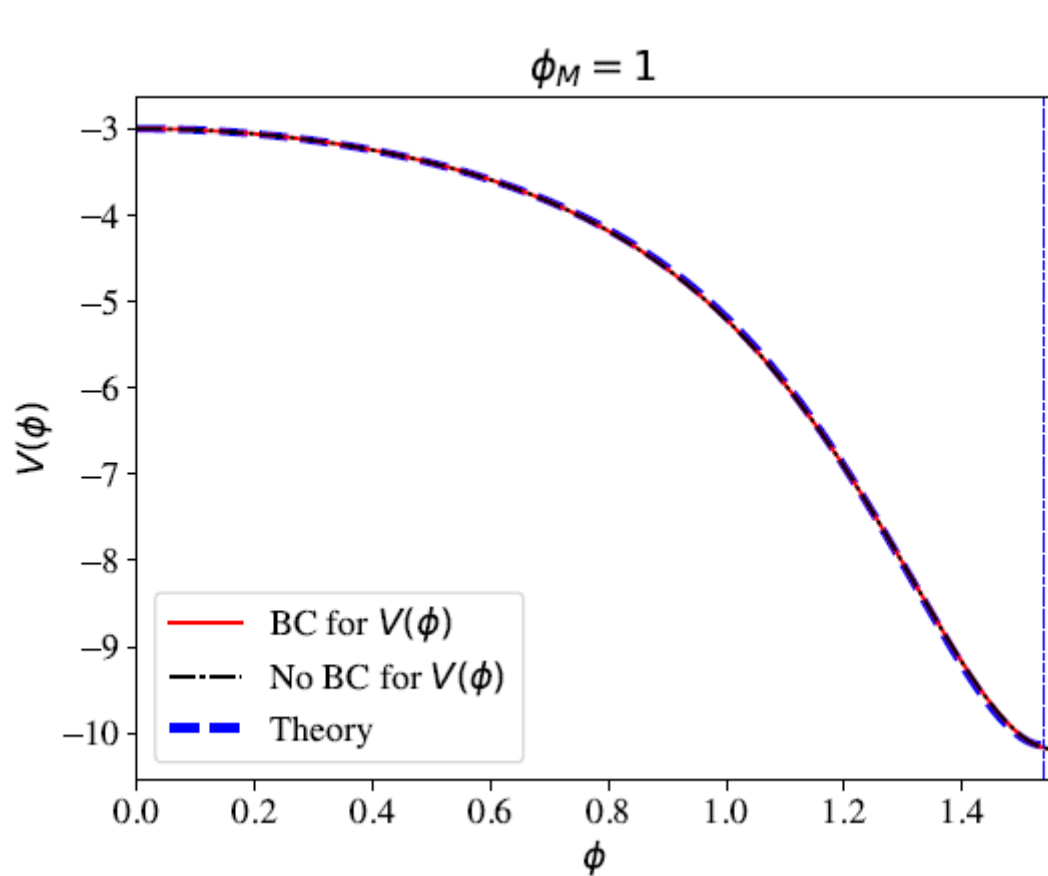
$$C_0 \equiv V(0) + 3 = 0, \quad C_1 \equiv V'(0) = 0, \quad C_2 \equiv V''(0) + 3 = 0.$$

Asymp. AdS

UV fixed point in the
Gauge theory

Fixes the conf. dimension
of the operator

POTENTIAL BOUNDARY CONDITIONS

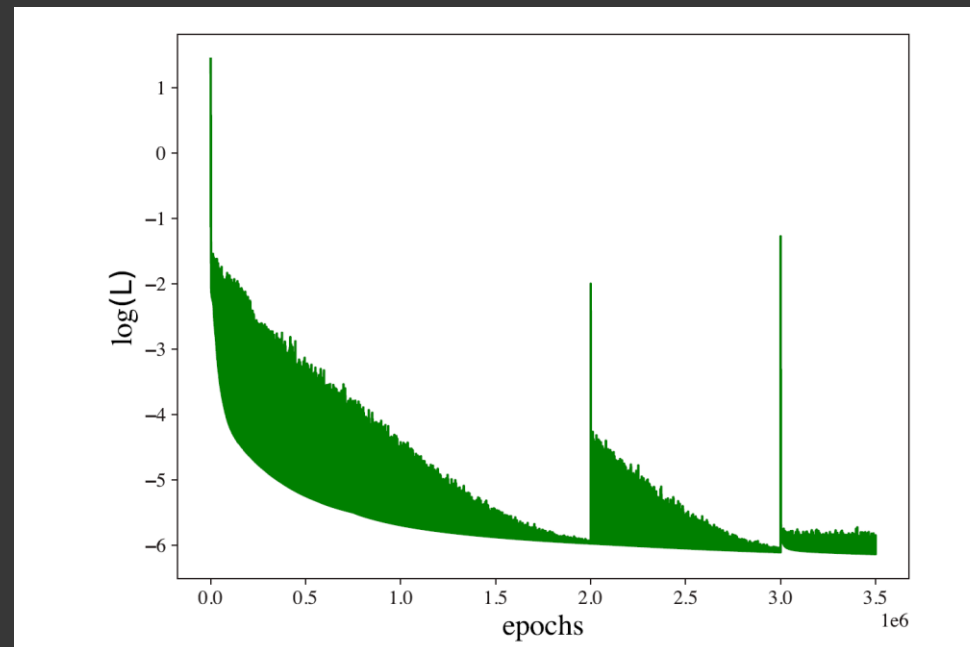


SAMPLING TECHNIQUES FOR THE NN

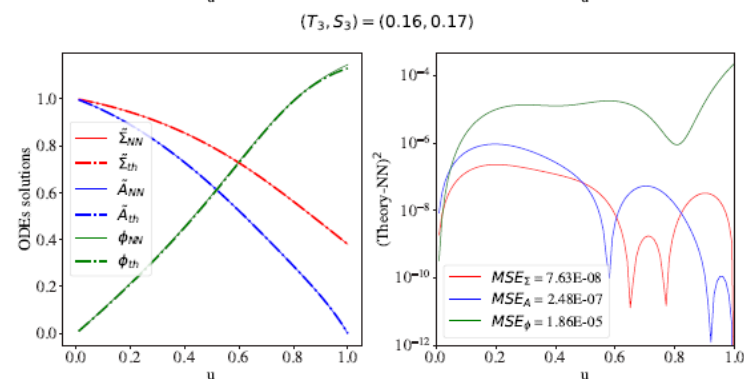
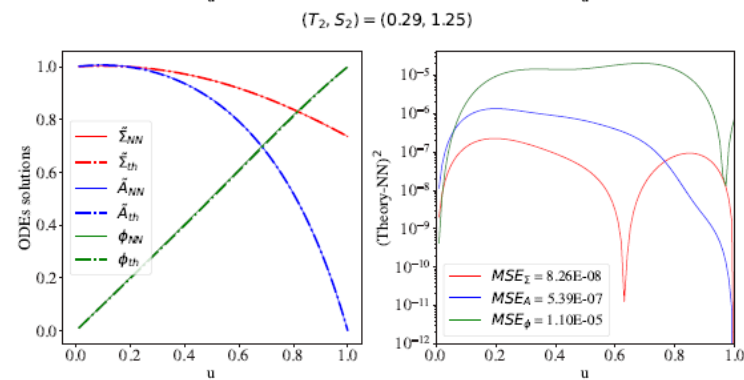
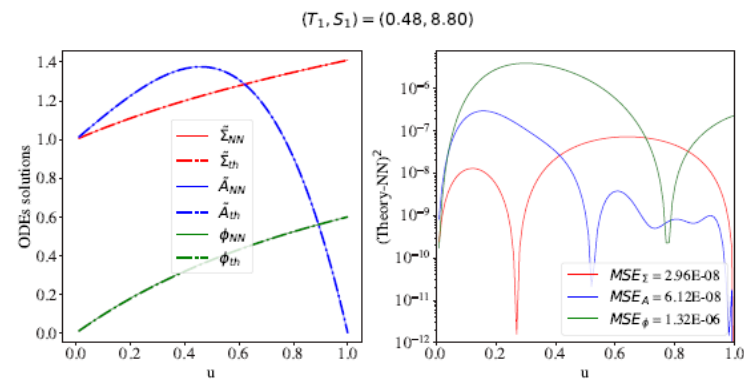
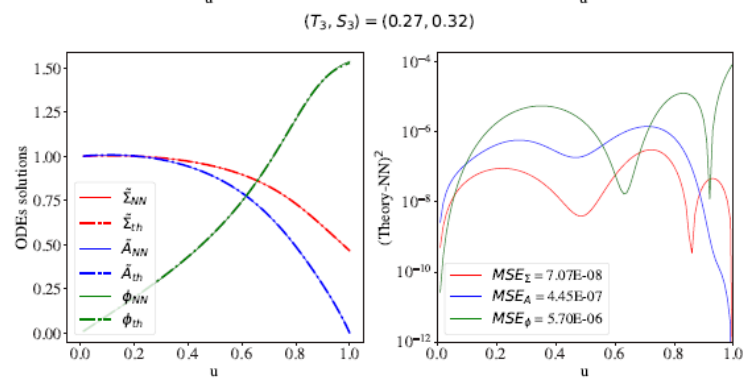
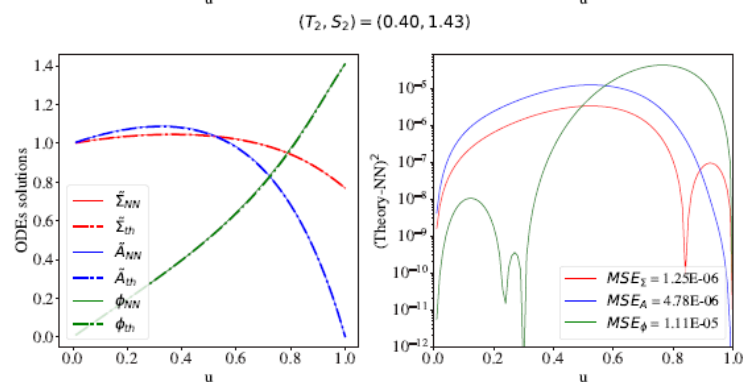
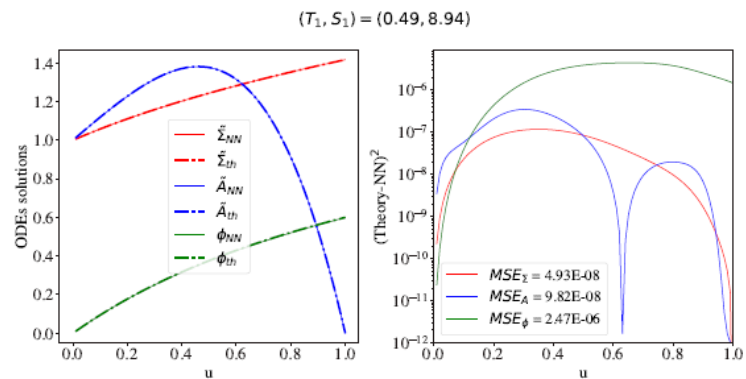
- The independent variable is sampled using Chebyshev nodes of the second kind. We have used 48 points. This allow us to better constrain the function near the boundaries of the interval.
- The sampling along the $S(T)$ is chosen to have in between 65-70 points.
- This leaves us with a batch size of approx. 3000 points.

TRAINING DETAILS

- 10 realizations of the same case trained 1e6 epochs.
- The case with the lowest loss is trained until the results are good enough.
- We add the b.c. of the potential as an additional loss and keep training for some epochs until they are fulfilled.



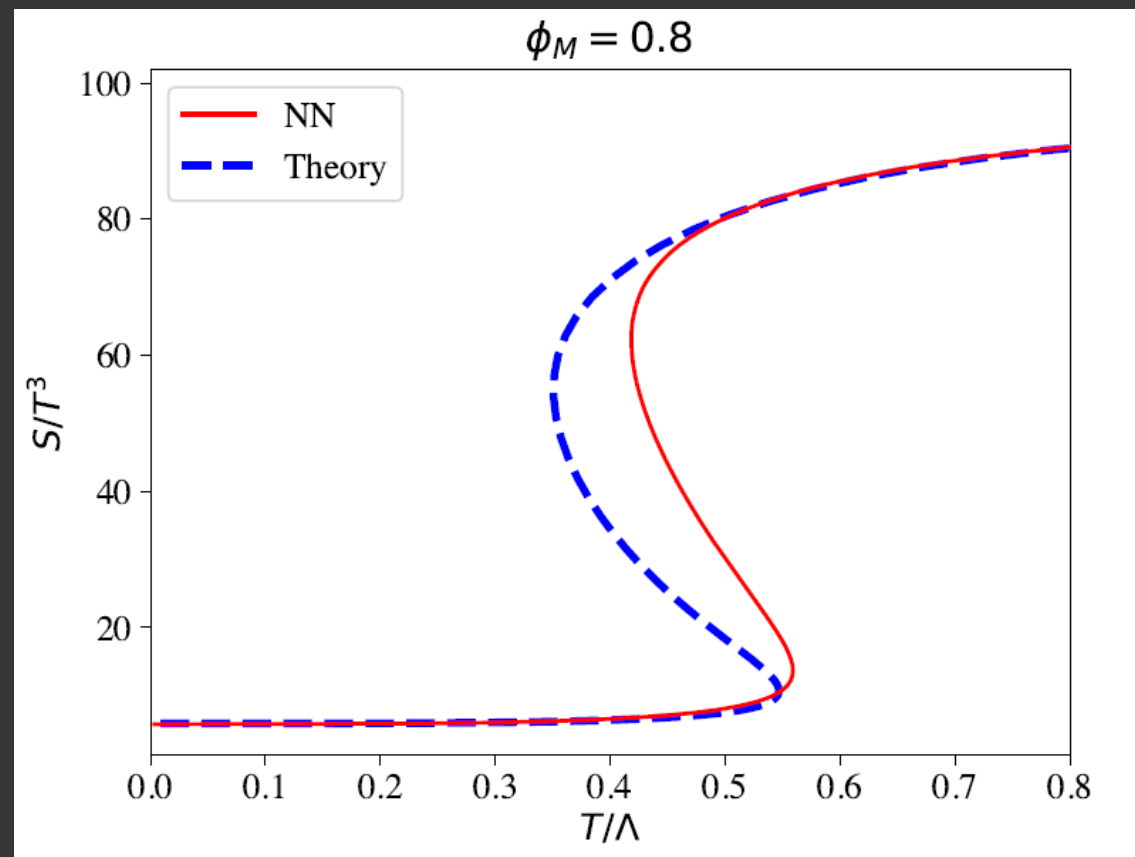
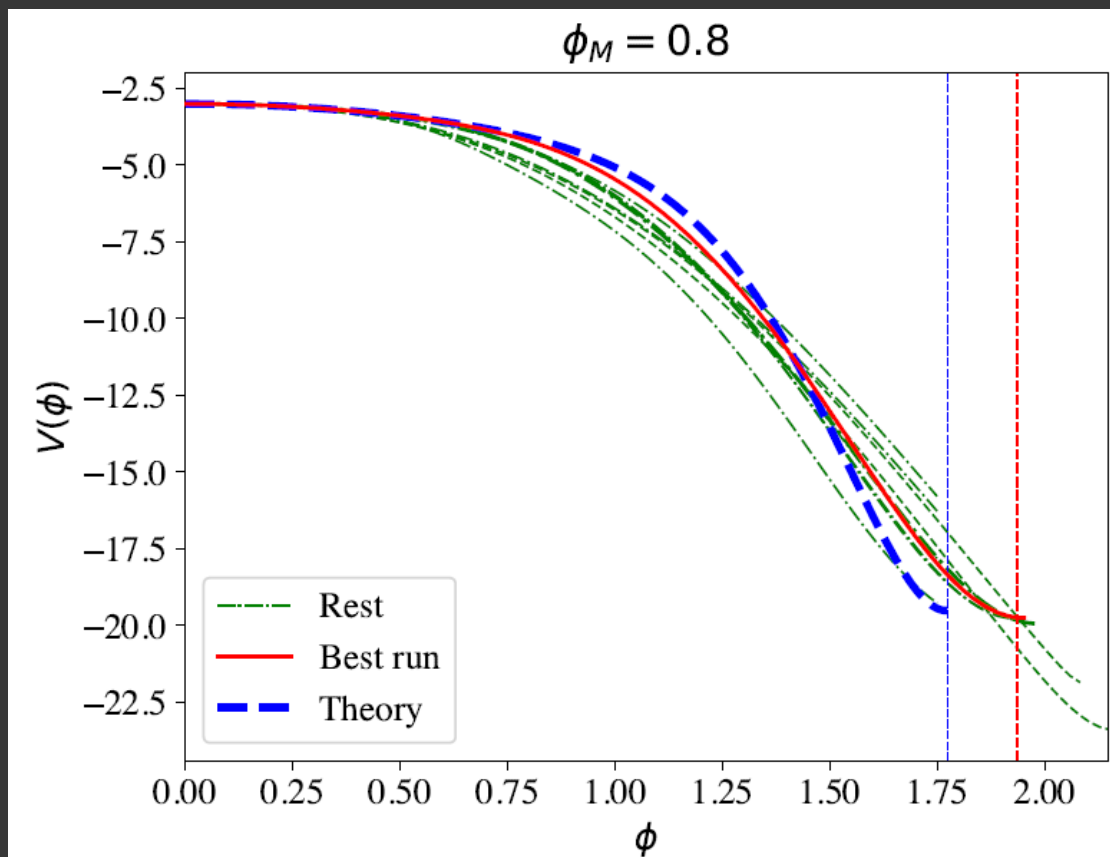
ODE SOLUTIONS



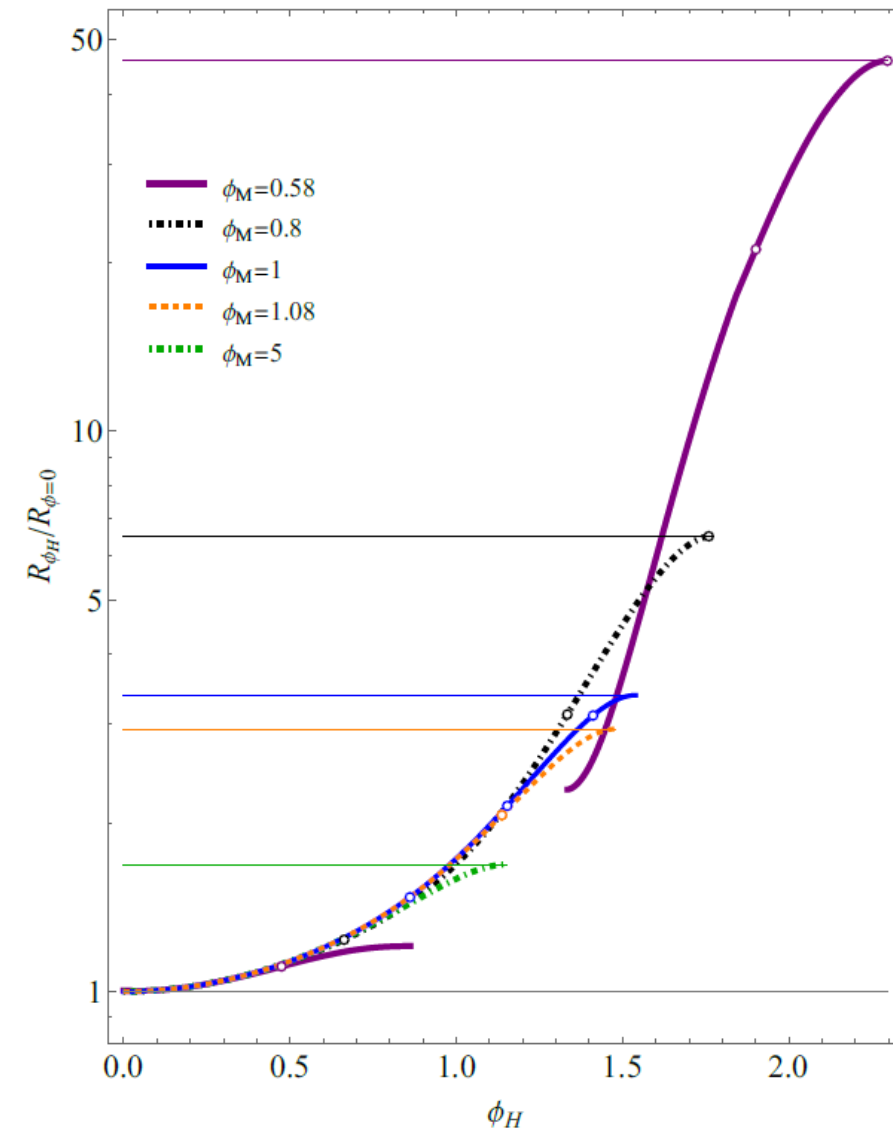
$$\phi_M = 1.0$$

$$\phi_M = 5.0$$

PATHOLOGICAL CASES



MULTI-SCALE OF THE PROBLEM



MULTIHEAD APPROACH

